Artificial Intelligence and Epistemic Resilience: A Theory of Collective Learning

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Abstract

Artificial intelligence (AI) increasingly shapes how people interpret information. Retraining on the informational environments it helps create can weaken society's capacity to correct errors. We develop a theory of epistemic resilience, defined as society's capacity to recover from informational mistakes through collective learning. We link individual perception rules to aggregate learning efficiency and identify systems whose aggregate dynamics depend only on current distortion and learning efficiency. The theory characterizes the threshold separating current correction from amplification and conditions for local recovery to a distortion-free steady state. An explicit retraining model shows how distorted training data can weaken collective learning. Under specified global conditions and with aggregate perceptions initially at their distortion-free values, sufficiently small disturbances disappear, whereas sufficiently large disturbances generate sustained amplification despite initially declining. Learning efficiency remains positive. Each current perception rule would support correct asymptotic learning if held fixed and applied to conditionally independent, identically distributed observations. Initial correction can therefore precede epistemic collapse as retraining weakens the capacity to correct subsequent errors. This result extends to nonlinear distortion multipliers under corresponding conditions. The reduced dynamics support comparisons across AI architectures and distinguish interventions changing current states from those changing subsequent dynamics. At a common initial state and fixed correction parameter, changes to those dynamics operate through distortion amplification and the evolution of learning efficiency. These results motivate governance that protects collective learning alongside information quality.

Keywords: Artificial intelligence; Epistemic resilience; Collective learning; Social learning; AI governance

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1 Introduction

Human societies constantly make mistakes. Markets misprice assets, scientists pursue incorrect hypotheses, citizens adopt inaccurate beliefs, and experts make erroneous predictions. Observation, communication, experimentation, and debate can expose these errors and lead people to revise their judgments. Science, markets, and democratic institutions depend in part on such processes of correction. Their effectiveness depends on how people interpret evidence and interact with one another, a concern also studied in research on collective intelligence (Woolley et al., 2010).

The capacity to correct errors can itself vary and change over time. Some errors are quickly identified, while others persist or accumulate. We call society's capacity to recover from informational mistakes through collective learning *epistemic resilience*. This concept draws on resilience as a property of dynamical systems (Holling, 1973, Walker et al., 2004).[1] Individuals contribute to this capacity by interpreting evidence, exchanging information, and challenging one another's conclusions. Independent interpretations can reveal errors that a shared interpretation overlooks.

Artificial intelligence (AI) can change these processes. AI systems help people search for information, summarize documents, rank sources, and interpret evidence. Their explanations can influence the rules people use to evaluate subsequent observations. Later retraining may draw on the informational environment shaped by these interactions, including online discussions, scientific summaries, educational material, and AI-assisted content. Evidence suggests that generative AI can alter participation and content production in online knowledge communities (Burtch et al., 2024). Research also shows that recursive training on AI-generated content can degrade model performance (Shumailov et al., 2024). When information shaped by AI enters subsequent training, changes in human interpretation and changes in AI behavior can reinforce one another. This feedback raises a question about *how society learns* and whether it retains the capacity to correct its mistakes.

Research on AI increasingly examines its interactions with human behavior, decision-making, and knowledge production (e.g., Bail, 2024, Feuerriegel et al., 2024, Goergen et al., 2025, Jackson et al., 2025). Theoretical work studies AI's effects on information aggregation, human cognition and learning incentives, and informational feedback between AI and society (Acemoglu, Kong and Ozdaglar, 2026, Acemoglu, Lin, Ozdaglar and Siderius, 2026, Bos and Bosi, 2026). These questions also connect to the literature on communication, social networks, and collective learning (e.g., Acemoglu et al., 2011, Dasaratha and He, 2020). We examine how perception rules evolve through social communication and AI retraining, and how this evolution changes collective error correction. People may continue receiving evidence and updating their beliefs while the rules used to interpret that evidence become less effective at distinguishing correct from incorrect conclusions.

The risk is that informational errors weaken the learning processes needed to correct them. This concern extends beyond misinformation generated directly by AI.[2] Social interaction can contribute to the spread and persistence of misinformation (Vicario et al., 2016). Retraining introduces a further possibility when the resulting information changes how future evidence is interpreted. A loss of correction capacity could then make errors arising in human communication, scientific research, financial markets, or political discourse harder to eliminate. The source of an error need not determine the consequences of its interaction with learning.

[1] Our notion of epistemic resilience also relates to social epistemology, which studies the collective production of knowledge (e.g., Kitcher, 1993, Goldman, 1999).

[2] Related research examines the linguistic features and detection of AI-generated misinformation (Ma et al., 2025), as well as interventions against online misinformation (Kozyreva et al., 2024).

Several scenarios illustrate this concern and motivate the formal analysis. Consider students using large language models to explain mathematics, programming, or scientific concepts. An explanation may help them solve a current problem. Their subsequent learning also depends on the interpretations they adopt from that explanation. If repeated reliance encourages deference to a mistaken interpretation without examining the reasoning, students may become less able to recognize corrective evidence in later problems.

In scientific research, AI systems can assist with literature reviews, references, code, and hypotheses. These tools can reduce the cost of processing information. Retraining on scientific outputs, including AI-assisted outputs, may also reinforce prevailing interpretations. If these interpretations discount contradictory evidence, researchers could receive more information while becoming less effective at revising an incorrect hypothesis.

Financial analysis raises a related concern. AI-generated summaries and forecasts may help analysts process news. Repeated reliance on similar interpretations, followed by retraining on AI-assisted reports, could make their assessments more alike. This becomes a risk to correction if the shared interpretation makes evidence against prevailing views harder to recognize. Markets could then respond quickly to news while remaining slow to revise those views.

In public debate, AI-generated explanations and recommendations can influence which sources and interpretations people encounter. Citizens may receive information from several sources while repeatedly encountering similar explanations. If those explanations make corrective evidence harder to interpret, collective learning can weaken even without an immediate increase in misinformation.

These scenarios concern changes in the capacity to evaluate future evidence. Their consequences depend on the perception rules that AI and social interaction produce. Greater reliance on AI or greater similarity of interpretations does not, by itself, establish a loss of learning efficiency. The formal analysis identifies when changes in interpretation can weaken correction enough to generate sustained amplification of informational errors.

We develop a theory of epistemic resilience to study this feedback. We model the perception rules through which agents interpret evidence and aggregate them using weights that reflect their long-run influence in the social-information network. We evaluate the resulting rule using the Frick–Iijima–Ishii learning-efficiency index (Frick et al., 2024). The index evaluates learning under a fixed perception rule. Applying it to the current rule makes aggregate learning efficiency an endogenous state variable as perceptions evolve. We focus on systems for which current informational distortion and aggregate learning efficiency suffice to determine next-period aggregate learning efficiency. We call these systems admissible. Together with the specified distortion law, this restriction gives a representation of their aggregate dynamics in two state variables. It identifies the current states and subsequent dynamics through which AI architectures and policy interventions affect epistemic resilience.

The general theory characterizes the learning-efficiency threshold separating correction from amplification of current distortions and the conditions for local return to a distortion-free steady state. We use *epistemic collapse* to describe a state in which positive informational distortions are amplified over the next period. This definition concerns correction at a given date. Subsequent recovery depends on how learning efficiency evolves.

We then specify an explicit retraining model in which informational distortion changes which observations are selected for training. Population likelihood fitting determines the AI perception rule, which enters individual perception updating. For the specified training distributions and selection rule, greater distortion lowers next-period aggregate learning efficiency at a given current efficiency. The construction derives

this response and the scalar epistemic transition while allowing heterogeneous individual perceptions. The distortion law remains an assumption of the model. Appendix A, Section 4.2, provides the construction.

This model shows how initial correction can be followed by sustained amplification. Under the stated global conditions and with initial aggregate perceptions fixed at their distortion-free values, sufficiently small disturbances disappear. Sufficiently large disturbances eventually amplify at every date and grow without bound along the autonomous trajectory, even though every positive disturbance initially declines (Theorem 2 in Appendix B). The amplification and correction parameters remain fixed. The reversal from correction to amplification is generated by the endogenous deterioration of learning efficiency. Efficiency remains strictly positive, and every current perception rule would support correct asymptotic learning if held fixed and applied to conditionally independent, identically distributed observations. The conclusion extends to positive, strictly decreasing nonlinear distortion multipliers under corresponding conditions (Proposition 4). Initial improvement in information quality can therefore coexist with erosion of the capacity to correct future errors. Sustained amplification under these dynamics does not rule out recovery following an intervention.

2 Theory

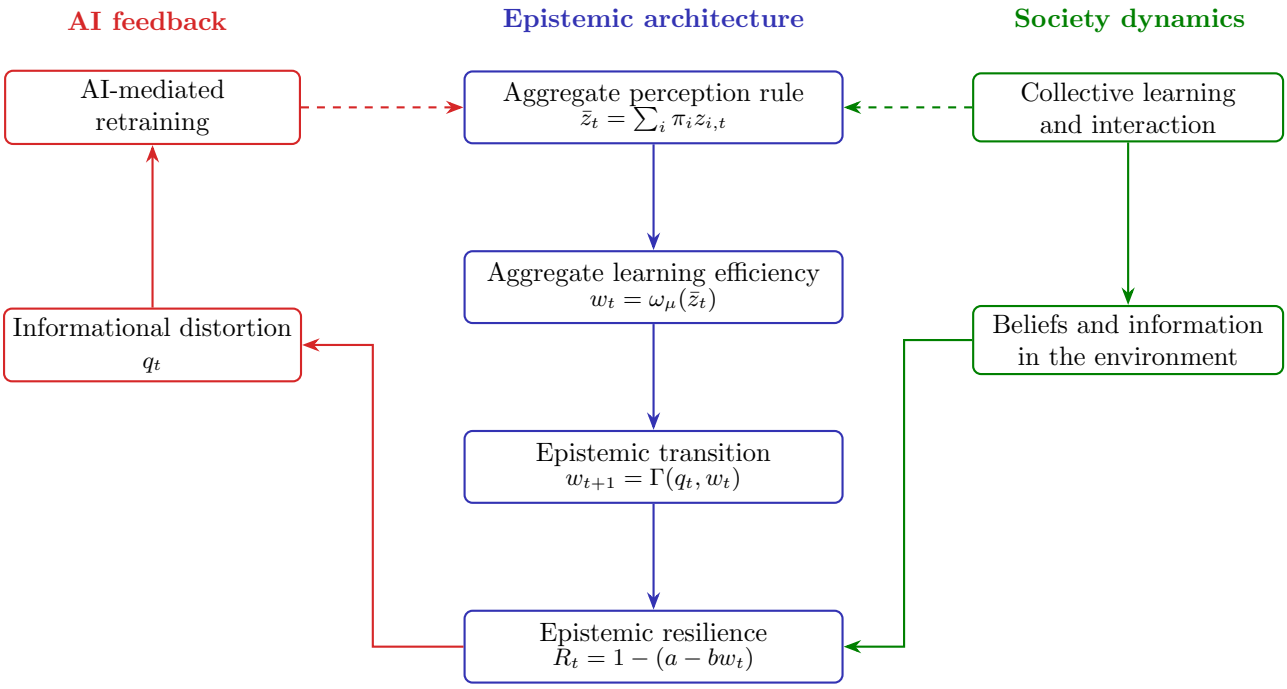


Figure 1: **Perception, retraining, and error correction.** Individual perception rules are aggregated using agents' long-run influence in the social-information network. The Frick-Iijima-Ishii index assigns a learning efficiency to the resulting rule. AI-mediated retraining changes perception rules using information from the environment that AI systems help shape. For admissible systems, current distortion and learning efficiency determine next-period learning efficiency through the epistemic transition. Current learning efficiency and the distortion law determine the resilience margin.

We study how AI-mediated retraining changes society's capacity to correct informational errors. We first define *epistemic resilience* and relate learning efficiency to individual perception rules. We then show

how social interaction and AI retraining change the aggregate perception rule and its learning efficiency. The reduced analysis applies to systems for which the current aggregate state suffices to determine its subsequent evolution, as defined below. Section 2.5 explains how an explicit retraining model can turn initial correction into sustained amplification after a sufficiently large disturbance. Appendix A provides the retraining construction. Appendix B states and proves the recovery and amplification result and its extension to a nonlinear distortion multiplier.

2.1 Defining epistemic resilience

Epistemic resilience describes the capacity of the AI–society learning system to correct informational errors. We describe correction using two aggregate variables, informational distortion and learning efficiency.

Let $q_t \geq 0$ denote aggregate informational distortion in the AI-mediated information environment at date t. Distortions may originate from AI outputs, human communication, scientific mistakes, financial misinterpretations, or political discourse. Their source is not essential. What matters is whether the collective learning process eliminates informational distortions or instead allows them to persist and accumulate over time.

Let $w_t \geq 0$ denote aggregate learning efficiency at date t. Sections 2.2 and 2.3 define this index from individual and aggregate perception rules. A higher w_t gives a smaller next-period distortion for a given positive current distortion. We define epistemic resilience as follows. Epistemic resilience concerns the dynamics of informational distortions rather than their level. The main question is whether small distortions disappear through collective learning or instead persist over time.

Definition 1 (Epistemic resilience)**.** *The AI–society learning system is epistemically resilient at date t if small informational distortions decrease over the next period. For a given current learning efficiency w_t, this means $q_{t+1} < q_t$ for all sufficiently small $q_t > 0$.*

We consider feasible states satisfying $a - bw_t \geq 0$, so that informational distortion remains nonnegative, and we assume that distortion evolves according to $q_{t+1} = (a - bw_t)q_t$. The parameter $a > 0$ captures the gross amplification of informational distortions through AI-mediated social communication, and $b > 0$ measures the effectiveness of learning efficiency in correcting distortions. For a given positive current distortion, higher learning efficiency reduces next-period distortion, whereas lower learning efficiency raises the distortion multiplier. Thus, epistemic resilience holds if and only if

$$a - bw_t < 1,$$

or equivalently $w_t > w^\star$ and $w^\star := \frac{a-1}{b}$. This threshold determines whether positive informational distortions decrease over the next period. If learning efficiency falls below the threshold, positive distortions are amplified.

Definition 2 (Comparative epistemic resilience)**.** *Consider two AI–society learning systems A and B, with distortion dynamics*

$$q^k_{t+1} = (a_k - b_k w^k_t)q^k_t, \quad k \in \{A, B\}.$$

System A is weakly more epistemically resilient than system B at date t if, for a common small positive distortion at that date, its next-period distortion is no greater:

$$a_A - b_A w^A_t \leq a_B - b_B w^B_t.$$

The comparison is strict if the inequality is strict.

Define the epistemic resilience margin by $R_t := 1 - (a - bw_t)$. The system is epistemically resilient at date t if and only if $R_t > 0$. For $q_t > 0$, the margin measures the proportional decrease in distortion over the next period. A larger margin means a greater reduction or weaker amplification. Whether distortion disappears over time also depends on the subsequent evolution of learning efficiency.

2.2 Perception rules and learning efficiency

We use the learning-efficiency index of Frick et al. (2024) to evaluate individual and aggregate perception rules at each date. It measures how quickly decision errors become unlikely under a fixed perception rule when that rule satisfies the consistency condition stated below.

Consider a binary state space $\Theta = \{\theta_L, \theta_H\}$ and a finite signal space X.[3] The true signal structure is

$$\mu = (\mu_{\theta_L}, \mu_{\theta_H}), \quad \mu_\theta \in \Delta(X),$$

where $\Delta(X)$ is the set of probability distributions on X. Both conditional distributions have full support, and $\mu_{\theta_L} \neq \mu_{\theta_H}$. Conditional on state θ, the observations used to evaluate a fixed perception rule are independent and identically distributed according to μ_θ.

At date t, agent i interprets signals through a perceived signal structure $\widehat{\mu}_{i,t} = (\widehat{\mu}_{i,t,\theta_L}, \widehat{\mu}_{i,t,\theta_H})$, also with distinct full-support conditional distributions. This perception rule specifies how signals provide evidence about the states. It is distinct from the agent's current belief about which state is true. Its log-likelihood-ratio vector is

$$z_{i,t} = \left(\log \frac{\widehat{\mu}_{i,t,\theta_H}(x)}{\widehat{\mu}_{i,t,\theta_L}(x)} \right)_{x \in X}.$$

The set of vectors generated by such perception rules is

$$\mathcal{Z} := \left\{ z \in \mathbb{R}^{|X|} : \min_{x \in X} z(x) < 0 < \max_{x \in X} z(x) \right\}$$

We call $\mathcal{Z}$ the realizability domain.[4]

For $\nu, \eta \in \Delta(X)$ with η of full support, let $D_{\mathrm{KL}}(\nu \| \eta) = \sum_{x \in X} \nu(x) \log(\nu(x)/\eta(x))$ denote the Kullback-Leibler divergence, with zero terms interpreted as zero. Following Frick et al. (2024), the empirical distributions under which the perceived states are indistinguishable form the set

$$\begin{aligned} \mathcal{I}(\widehat{\mu}) &= \{\nu \in \Delta(X) : D_{\mathrm{KL}}(\nu \| \widehat{\mu}_{\theta_L}) = D_{\mathrm{KL}}(\nu \| \widehat{\mu}_{\theta_H})\} \\ &= \{\nu \in \Delta(X) : \langle \nu, z(\widehat{\mu}) \rangle = 0\}. \end{aligned}$$

The equality follows from $D_{\mathrm{KL}}(\nu \| \widehat{\mu}_{\theta_L}) - D_{\mathrm{KL}}(\nu \| \widehat{\mu}_{\theta_H}) = \langle \nu, z(\widehat{\mu}) \rangle$. The learning-efficiency index therefore

[3] The binary-state restriction simplifies the representation but is not essential to the reduced theory of epistemic resilience. With more than two states, learning efficiency may be defined as the smallest exponential rate at which any incorrect state is selected in place of the true state, or represented by a collection of state-pair-specific rates. Provided that the underlying perception and retraining dynamics are admissible and induce an aggregate learning-efficiency transition $w_{t+1} = \Gamma_{\mathcal{A}}(q_t, w_t)$, the resilience characterization below is unchanged.

[4] A vector $z \in \mathbb{R}^{|X|}$ is the log-likelihood-ratio vector of a pair of distinct full-support probability distributions if and only if it has at least one strictly positive and one strictly negative component. Indeed, if $z(x) = \log(\widehat{\mu}_{\theta_H}(x)/\widehat{\mu}_{\theta_L}(x))$, normalization of the two probability distributions implies that z must take both signs. Conversely, both signs imply $\min_x e^{z(x)} < 1 < \max_x e^{z(x)}$. One can therefore choose a full-support probability distribution $\widehat{\mu}_{\theta_L}$ satisfying $\sum_x \widehat{\mu}_{\theta_L}(x) e^{z(x)} = 1$ and define $\widehat{\mu}_{\theta_H}(x) = \widehat{\mu}_{\theta_L}(x) e^{z(x)}$.

depends on $\widehat{\mu}$ only through $z(\widehat{\mu})$. For every $z \in \mathcal{Z}$, define

$$\omega_\mu(z) := \min_{\theta \in \Theta} \min_{\substack{\nu \in \Delta(X) \\ \langle \nu, z \rangle = 0}} D_{\mathrm{KL}}(\nu \| \mu_\theta).$$

Thus, for every full-support perceived signal structure with distinct conditional distributions, $w(\mu, \widehat{\mu}) = \omega_\mu(z(\widehat{\mu}))$. Each inner minimization gives the KL projection of μ_θ onto the indistinguishability set. The index is smaller when this set lies closer, in KL divergence, to one of the true signal distributions. Its interpretation as a rate of error correction requires the following consistency condition.

A realizable perception rule need not lead to correct asymptotic learning. For the error-correction interpretation, we restrict perception rules to the consistency domain

$$\mathcal{C}_\mu := \{ z \in \mathcal{Z} : \langle \mu_{\theta_L}, z \rangle < 0 < \langle \mu_{\theta_H}, z \rangle \}.$$

This is the log-likelihood-ratio formulation of the consistency condition in Frick et al. (2024). For a perceived signal structure generating z, it is equivalent to

$$D_{\mathrm{KL}}(\mu_\theta \| \widehat{\mu}_\theta) < D_{\mathrm{KL}}(\mu_\theta \| \widehat{\mu}_{\theta'}) \text{ for every } \theta \neq \theta'.$$

The condition requires the expected perceived log-likelihood ratio to favor the true state. Hold the perception rule fixed and take a prior assigning positive probability to both states. Under the conditionally independent and identically distributed observations specified above, Bayesian updating using that rule assigns probability converging to one to the true state almost surely. Consistency allows $\widehat{\mu} \neq \mu$, and $\mathcal{C}_\mu$ is nonempty because it contains the true log-likelihood-ratio vector.

For each current consistent perception rule, let us define agent i's instantaneous learning efficiency by

$$w_{i,t} := w(\mu, \widehat{\mu}_{i,t}) = \omega_\mu(z_{i,t}).$$

Here *instantaneous* refers to an evaluation at date t. We hold the current perception rule fixed while the sample size increases. Under consistency, $w_{i,t}$ is the exponential decay rate of the worst-state error probability for the fixed binary decision rule described in Appendix B, Section 5.1. It is not the error exponent of the entire evolving AI–society process.

AI-mediated retraining changes perception rules and makes their learning efficiencies endogenous. Section 2.3 defines the aggregate index and derives its evolution from perception updating. A decline in aggregate learning efficiency weakens correction and leads to amplification of positive distortions if w_t falls below $w^\star$.

2.3 AI-mediated perception dynamics

Society consists of agents $i = 1, \ldots, n$ connected through a social-information network represented by an irreducible row-stochastic matrix

$$S = (s_{ij}) \text{ with } s_{ij} \geq 0 \text{ and } \sum_j s_{ij} = 1.$$

The entry s_{ij} is the weight that agent i assigns to agent j's perception rule when updating her own interpretation of signals. Since S is irreducible and row-stochastic, it admits a unique strictly positive stationary

distribution π, satisfying

$$\pi^\top S = \pi^\top \text{ with } \pi^\top \mathbf{1} = 1,$$

and where $\mathbf{1}$ denotes the vector of ones. We interpret π_i as the long-run influence weight of agent i in the network. Although perception evolves at the individual level, epistemic resilience is a property of the collective learning process rather than of any particular agent. We therefore define the aggregate perception rule by

$$\bar{z}_t := \sum_{i=1}^{n} \pi_i z_{i,t}.$$

This stationary network-weighted average gives greater weight to agents with greater long-run network influence. Individual perception rules are restricted to $\mathcal{C}_\mu$. Their aggregate also satisfies the consistency inequalities, because

$$\langle \mu_{\theta_L}, \bar{z}_t \rangle = \sum_{i=1}^{n} \pi_i \langle \mu_{\theta_L}, z_{i,t} \rangle < 0 < \sum_{i=1}^{n} \pi_i \langle \mu_{\theta_H}, z_{i,t} \rangle = \langle \mu_{\theta_H}, \bar{z}_t \rangle.$$

Hence $\bar{z}_t \in \mathcal{C}_\mu \subseteq \mathcal{Z}$. It corresponds to a perceived signal structure with distinct full-support conditional distributions, satisfies consistency, and has a nonempty indistinguishability set. We define aggregate learning efficiency by

$$w_t := \omega_\mu(\bar{z}_t). \tag{1}$$

This index evaluates the aggregate perception rule. In general, it differs from the influence-weighted average of individual learning efficiencies,

$$\omega_\mu \left(\sum_{i=1}^{n} \pi_i z_{i,t} \right) \neq \sum_{i=1}^{n} \pi_i \omega_\mu(z_{i,t}).$$

The distinction arises because the Frick-Iijima-Ishii index is a nonlinear mapping from perception rules to learning efficiency. Aggregate learning efficiency is therefore obtained by first aggregating perception rules according to their long-run network influence and then evaluating the resulting aggregate perception rule.

The aggregate index has a precise interpretation when the current rule is held fixed.[5] Fix a date t and the current perception profile. Consider an auxiliary sample $Y_1, \ldots, Y_m$ drawn independently from μ_θ conditional on state θ. The influence weighted interpretation of this common sample is the score

$$L_{t,m} := \sum_{i=1}^{n} \pi_i \sum_{k=1}^{m} z_{i,t}(Y_k) = \sum_{k=1}^{m} \bar{z}_t(Y_k).$$

Let $\widehat{\theta}_{t,m}$ select θ_H when $L_{t,m} \geq 0$ and θ_L otherwise. Since $\bar{z}_t \in \mathcal{C}_\mu$, its worst-state error probability

$$\varepsilon_{t,m} := \max_{\theta \in \Theta} \mathbb{P}_\theta \left(\widehat{\theta}_{t,m} \neq \theta \right)$$

satisfies

$$\lim_{m \to \infty} -\frac{1}{m} \log \varepsilon_{t,m} = \omega_\mu(\bar{z}_t) = w_t.$$

Here m is the sample size, not calendar time. This auxiliary decision rule gives the chosen aggregate index an operational interpretation. It does not require agents in the model to observe a common sample, nor does

[5] Appendix B, Section 5.1, proves the error-exponent identity used here.

it establish the error exponent of an arbitrary collective decision procedure or of the evolving AI–society process.

At date t, AI-mediated interpretation is represented by a perceived signal structure $\widehat{\mu}_t^A = \left(\widehat{\mu}_{t,\theta_L}^A, \widehat{\mu}_{t,\theta_H}^A\right)$ with distinct full-support conditional distributions. Its log-likelihood-ratio vector z_t^A is defined in the same way as $z_{i,t}$ in Section 2.2. It describes the interpretation of signals conveyed through AI-mediated explanations, summaries, rankings, and recommendations. Agents update their perceived log-likelihood ratios according to

$$z_{i,t+1} = (1-\lambda) z_{i,t} + \lambda \left[(1-\rho) \sum_{j=1}^{n} s_{ij} z_{j,t} + \rho z_t^A \right], \tag{2}$$

where $\lambda \in (0,1]$ is the speed of perception adjustment and $\rho \in [0,1]$ measures reliance on AI relative to social interpretation. The update combines the agent's previous rule with the rules supplied by the social network and AI. Within the adjustment component, increasing ρ shifts weight from social interpretation to AI.

OB: is it more clear if we say that here instead (or is it a repeatition somehow?): Equation (2) captures a simple behavioral mechanism. Agents revise their interpretation of future information by combining three sources: their previous perception rule, the perception rules prevailing in their social environment, and the perception rule conveyed through AI-mediated explanations, summaries, rankings, and recommendations. The parameter ρ therefore measures the extent to which AI-mediated interpretation replaces social influence in shaping how future information is interpreted.

We initialize individual rules in $\mathcal{C}_\mu$ and require $z_t^A \in \mathcal{C}_\mu$ at every date along the paths considered. These conditions preserve consistency under the perception dynamics. The two strict expectation inequalities defining $\mathcal{C}_\mu$ are preserved by convex combinations and themselves imply that a vector has both positive and negative components. Thus, $\mathcal{C}_\mu$ is convex. Since the coefficients in Equation (2) are nonnegative and sum to one,

$$z_{i,t} \in \mathcal{C}_\mu \text{ for every } i, \quad z_t^A \in \mathcal{C}_\mu \implies z_{i,t+1} \in \mathcal{C}_\mu \text{ for every } i.$$

By induction, all individual rules remain in $\mathcal{C}_\mu$, as does their aggregate. Updating and aggregation therefore preserve both realizability and the interpretation of learning efficiency in terms of errors under a fixed rule. Applying ω_μ to Equation (2) gives the individual learning-efficiency update,

$$w_{i,t+1} = \omega_\mu \left((1-\lambda) z_{i,t} + \lambda \left[(1-\rho) \sum_{j=1}^{n} s_{ij} z_{j,t} + \rho z_t^A \right] \right).$$

Individual learning efficiency is thus determined by the updated perception rule. It is obtained by applying the Frick-Iijima-Ishii index to the perception rule generated by the underlying updating process. Weighting Equation (2) by π_i and summing across agents gives the aggregate perception dynamics,

$$\bar{z}_{t+1} = (1-\lambda\rho)\bar{z}_t + \lambda\rho z_t^A. \tag{3}$$

Indeed,

$$\begin{aligned}\bar{z}_{t+1} &= (1-\lambda)\sum_{i=1}^{n}\pi_i z_{i,t} + \lambda(1-\rho)\sum_{i=1}^{n}\pi_i\sum_{j=1}^{n}s_{ij}z_{j,t} + \lambda\rho\sum_{i=1}^{n}\pi_i z_t^A \\ &= (1-\lambda)\bar{z}_t + \lambda(1-\rho)\bar{z}_t + \lambda\rho z_t^A \\ &= (1-\lambda\rho)\bar{z}_t + \lambda\rho z_t^A.\end{aligned}$$

The second equality uses $\pi^\top S = \pi^\top$ and $\sum_i \pi_i = 1$. By the definition of w_t, Equation (3) then yields

$$w_{t+1} = \omega_\mu\left((1-\lambda\rho)\bar{z}_t + \lambda\rho z_t^A\right). \tag{4}$$

This is the exact aggregate learning-efficiency update implied by Equation (2) and our aggregation rule. It applies the learning-efficiency (Frick-Iijima-Ishii) index to the updated aggregate perception rule, not to an average of individual learning efficiencies.

Social averaging preserves the π-weighted mean of perception rules, so the detailed entries of S cancel from the aggregate recursion. For fixed λ and ρ, networks with the same stationary distribution generate identical aggregate perception and learning-efficiency paths if they have the same initial aggregate rule and the same AI perception path $(z_t^A)_{t\geq 0}$. This invariance is conditional on the AI path. Network structure can change the distribution of individual perception rules and learning efficiencies. Retraining may respond to these differences, so networks with the same π need not generate the same AI perception path. They can therefore have different aggregate dynamics in the full feedback system. The AI perception rule is itself generated by retraining on the informational environment that the AI system helps create. To formalize this, let

$$\boldsymbol{z}_t := (z_{1,t}, \ldots, z_{n,t})$$

denote the profile of human perception rules, and let r_t denote the internal retraining state of the AI system.[6] This state may record previous training data, model parameters, or other system-specific information inherited from earlier periods. An AI-mediated perception system $\mathcal{A}$ has a retraining operator

$$z_t^A = \mathcal{H}_{\mathcal{A}}(q_t, \boldsymbol{z}_t, r_t), \tag{5}$$

whose values belong to $\mathcal{C}_\mu$ along the paths considered, as required by the consistency requirement stated above. If retraining has no additional internal state, r_t may be omitted.

The operator $\mathcal{H}_{\mathcal{A}}$ combines three steps: current distortion, human perception rules, and the internal state determine the retraining environment. Retraining on that environment produces a perceived signal structure $\widehat{\mu}_t^A$, from which the log-likelihood-ratio vector z_t^A is obtained. The model works directly with this composite operator because both Equation (2) and ω_μ depend on the AI perception rule only through z_t^A. Any full-support perceived signal structures generating the same vector are equivalent for these dynamics and learning-efficiency indices.

The scalar q_t measures informational distortion in the environment. The vector z_t^A describes how the AI system interprets signals. Retraining links the two by translating changes in the informational environment into changes in the AI perception rule. This link need not be one-to-one. The same distortion level can produce different AI rules depending on the human perception profile, the internal state, and the architecture and retraining procedure represented by $\mathcal{H}_{\mathcal{A}}$.

[6] The internal retraining state may itself evolve according to a system-specific transition $r_{t+1} = \mathcal{F}_{\mathcal{A}}(q_t, \boldsymbol{z}_t, r_t)$. Its form is left implicit because the general analysis specifies conditions on the induced aggregate learning-efficiency transition.

Substituting Equation (5) into Equation (4) gives

$$w_{t+1} = \omega_\mu\left((1-\lambda\rho)\bar{z}_t + \lambda\rho\mathcal{H}_\mathcal{A}(q_t, \boldsymbol{z}_t, r_t)\right). \tag{6}$$

This equation makes explicit how the informational environment, AI retraining, human perception dynamics, and aggregate learning efficiency are connected. Different AI architectures, datasets, and retraining procedures correspond to different operators $\mathcal{H}_\mathcal{A}$ and potentially different internal states. Equation 6 determines next-period learning efficiency from the perception and retraining state. It does not generally reduce to a transition in (q_t, w_t) alone. The full profile $\boldsymbol{z}_t$ and the internal state r_t may still matter when q_t and $w_t = \omega_\mu(\bar{z}_t)$ are held fixed. A scalar Markov transition

$$w_{t+1} = \Gamma_\mathcal{A}(q_t, w_t)$$

therefore requires an additional restriction on the underlying perception and retraining system, which we call admissibility.

The particular updating rule in Equation (2) is one way to generate an evolution of aggregate perception. The reduced analysis also applies to other perception and retraining dynamics that satisfy the admissibility and regularity conditions below. In each case, perception rules are specified first and learning efficiency is derived from them. To define admissibility, let $x_t := (\boldsymbol{z}_t, r_t)$ denote the perception and retraining state, and write

$$\mathcal{W}(x_t) := \omega_\mu(\bar{z}_t) = w_t$$

for the aggregate learning efficiency associated with that state. Given current informational distortion q_t and state x_t, Equation (6) defines the aggregate learning-efficiency transition

$$\Phi_\mathcal{A}(q_t, x_t) := \omega_\mu\left((1-\lambda\rho)\bar{z}_t + \lambda\rho\mathcal{H}_\mathcal{A}(q_t, \boldsymbol{z}_t, r_t)\right). \tag{7}$$

Thus, $w_{t+1} = \Phi_\mathcal{A}(q_t, x_t)$. The mapping $\Phi_\mathcal{A}$ combines retraining, individual perception updating, network aggregation, and the Frick-Iijima-Ishii index.

Definition 3 (Admissible AI-mediated perception system)**.** *An AI-mediated perception system $\mathcal{A}$ is admissible if current distortion and aggregate learning efficiency suffice to determine next-period aggregate learning efficiency. That is, there exists a mapping*[7] $\Gamma_\mathcal{A} : \mathbb{R}_+ \times \mathbb{R}_+ \longrightarrow \mathbb{R}_+$ *such that, for every state (q_t, x_t) along the paths considered,*

$$\Phi_\mathcal{A}(q_t, x_t) = \Gamma_\mathcal{A}\big(q_t, \mathcal{W}(x_t)\big).$$

Equivalently, $w_{t+1} = \Gamma_\mathcal{A}(q_t, w_t)$. We call $\Gamma_\mathcal{A}$ the reduced epistemic transition induced by the AI-mediated perception system $\mathcal{A}$.

Admissibility requires that, for any given level of informational distortion q, two states x and x' with the same aggregate learning efficiency must generate the same next-period aggregate learning efficiency, provided that both (q, x) and (q, x') are among the states considered. That is,

$$\mathcal{W}(x) = \mathcal{W}(x') \implies \Phi_\mathcal{A}(q, x) = \Phi_\mathcal{A}(q, x'). \tag{8}$$

[7] Only the values of $\Gamma_\mathcal{A}$ on reduced states reachable under system $\mathcal{A}$ affect the induced dynamics. Writing its domain as $\mathbb{R}^2_+$ is a convenient extension.The local analysis requires a representation that admits a continuously differentiable extension to a neighborhood of the distortion-free state considered below.

Individual perception rules, their distribution across the network, and the internal retraining state may differ. Admissibility requires that these differences have no additional effect on next-period aggregate learning efficiency once (q_t, w_t) is fixed. Past perception rules, AI outputs, and informational distortions can affect the current perception and retraining state. Under admissibility, their effect on next-period aggregate learning efficiency is summarized by (q_t, w_t). This pair need not describe the full evolution of individual perceptions or the internal retraining state. It is sufficient for the aggregate learning-efficiency dynamics studied here. Different AI architectures and retraining mechanisms can therefore generate different individual dynamics while inducing the same epistemic transition. The general analysis characterizes resilience through this transition, without specifying the internal computations of a particular AI architecture.

Let $\bar{w}$ denote aggregate learning efficiency at the distortion-free state. The next assumption requires $\bar{w}$ to be a locally attracting fixed point of the learning-efficiency update when distortion is held at zero. It also specifies how learning efficiency responds locally to distortion.

Assumption 1 (Regularity and distortion sensitivity of the epistemic transition)**.** *Let $\mathcal{A}$ be an admissible AI-mediated perception system. Its induced epistemic transition $\Gamma_{\mathcal{A}}$ is continuously differentiable*[8] *in a neighborhood of the distortion-free state $(0, \bar{w})$ and satisfies*

$$\Gamma_{\mathcal{A}}(0, \bar{w}) = \bar{w}, \quad |\Gamma_{\mathcal{A},w}(0, \bar{w})| < 1, \quad \Gamma_{\mathcal{A},q}(0, \bar{w}) < 0.$$

AI-mediated retraining makes aggregate learning efficiency endogenous by changing the perception rules used to interpret evidence. For admissible systems, this feedback is represented by $w_{t+1} = \Gamma_{\mathcal{A}}(q_t, w_t)$. The additional condition

$$\Gamma_{\mathcal{A},q}(0, \bar{w}) < 0$$

requires an increase in distortion from the distortion-free state to reduce next-period learning efficiency locally, holding current learning efficiency fixed. Under the distortion law of Section 2.1, lower learning efficiency weakens subsequent error correction. The negative response is a restriction on retraining systems, not a consequence of admissibility alone. Assumption 1 imposes local conditions and places no additional restrictions on the transition away from this state. The later recovery and amplification analysis establishes sustained amplification for an explicit retraining model under additional global conditions.

2.4 Characterizing epistemic resilience

For an admissible AI-mediated perception system, the distortion law and the induced learning-efficiency transition give the two-dimensional AI–society system

$$\begin{cases} q_{t+1} = (a - bw_t)q_t, \\ w_{t+1} = \Gamma_{\mathcal{A}}(q_t, w_t). \end{cases}$$

The first equation is the distortion law introduced in Section 2.1. It determines whether current distortions decrease or increase over the next period. The second equation follows from the perception and retraining dynamics under admissibility, and determines how current distortion and learning efficiency affect next-period learning efficiency. Assumption 1 makes $(0, \bar{w})$ a distortion-free steady state. The parameter a measures gross distortion amplification, and b measures the effectiveness of correction at a given learning efficiency.

[8] Subscripts denote partial derivatives.

The transition $\Gamma_{\mathcal{A}}$ describes how learning efficiency changes over time. Together with the reduced initial state, these objects determine the reduced trajectory and its resilience margins.

Proposition 1 (Reduced representation and epistemic equivalence)**.** *Let $\mathcal{A}$ be an admissible AI-mediated perception system. Its reduced dynamics are represented by the triplet $(a, b, \Gamma_{\mathcal{A}})$. If two admissible systems $\mathcal{A}$ and $\mathcal{B}$ have the same parameters a and b and satisfy*

$$\Gamma_{\mathcal{A}} = \Gamma_{\mathcal{B}}$$

on a common feasible reduced state space that is forward invariant under both systems, they generate identical trajectories (q_t, w_t) from every common reduced initial condition (q_0, w_0) in that space. They therefore have identical resilience margins and epistemic-collapse thresholds. At any common distortion-free steady state, they also have the same local stability properties relative to that state space.

AI architectures can differ in their internal organization and retraining procedures while inducing the same reduced dynamics. Proposition 1 identifies when these differences leave the evolution of collective error correction unchanged. It allows comparisons through (a, b, Γ) without separately describing each architecture's internal computations. This observation also shifts the object of analysis from AI architectures themselves to the reduced dynamical objects governing collective learning.

Corollary 1 (Policy decomposition)**.** *Within the class of admissible AI-mediated perception systems, hold the post-intervention reduced initial condition fixed. Every policy intervention affecting epistemic resilience must modify at least one component of (a, b, Γ). If the distortion-correction parameter b is also fixed, every such intervention operates through one or both of the following components:*

(i) The distortion-amplification parameter a.

(ii) The induced epistemic transition Γ.

Interventions can also change the current reduced state. For example, a feasible intervention at date τ that changes w_τ to $\widetilde{w}_\tau$, holding q_τ and (a, b, Γ) fixed, changes the current resilience margin by

$$\widetilde{R}_\tau - R_\tau = b(\widetilde{w}_\tau - w_\tau),$$

without changing the collapse threshold. Reducing q_τ alone leaves the current margin unchanged but can affect subsequent distortion and learning efficiency. These changes in the current state are distinct from the changes in the transition laws compared in Corollary 1.

Epistemic resilience concerns correction over one period. Conditional on current learning efficiency, a small positive distortion is self-correcting if it decreases at the next date. Local asymptotic stability concerns the joint return of distortion and learning efficiency to $(0, \bar{w})$ after a sufficiently small feasible perturbation. The following theorem characterizes the first property. Under the conditions of Corollary 2 in Appendix B, Section 5.5, the same strict inequality also characterizes local asymptotic stability of the reduced system.

Theorem 1 (Characterization of epistemic resilience)**.** *Suppose Assumption 1 holds. At and locally around the distortion-free steady state $(0, \bar{w})$, define the resilience margin*

$$\bar{R} := 1 - (a - b\bar{w}).$$

The system is locally epistemically resilient if and only if $\bar{R} > 0$*, or equivalently* $\bar{w} > w^\star = \frac{a-1}{b}$*. Moreover, for any two systems* A *and* B*, system* A *is locally more epistemically resilient than system* B *if and only if* $\bar{R}_A \geq \bar{R}_B$*, that is,* $1 - (a_A - b_A\bar{w}_A) \geq 1 - (a_B - b_B\bar{w}_B)$*.*

Theorem 1 shows that $\bar{R}$ summarizes local correction at the distortion-free steady state, as a sufficient statistic for local epistemic resilience. Along the dynamic path, the corresponding instantaneous resilience margin is $R_t := 1 - (a - bw_t)$.

For a common positive current distortion, a larger margin means stronger correction or weaker amplification over the next period. Perception updating and retraining change this margin through w_t, while a and b determine how learning efficiency translates into correction. Thus, systems can be ranked by their current resilience margins without separately comparing the architectures that generate their perception dynamics.

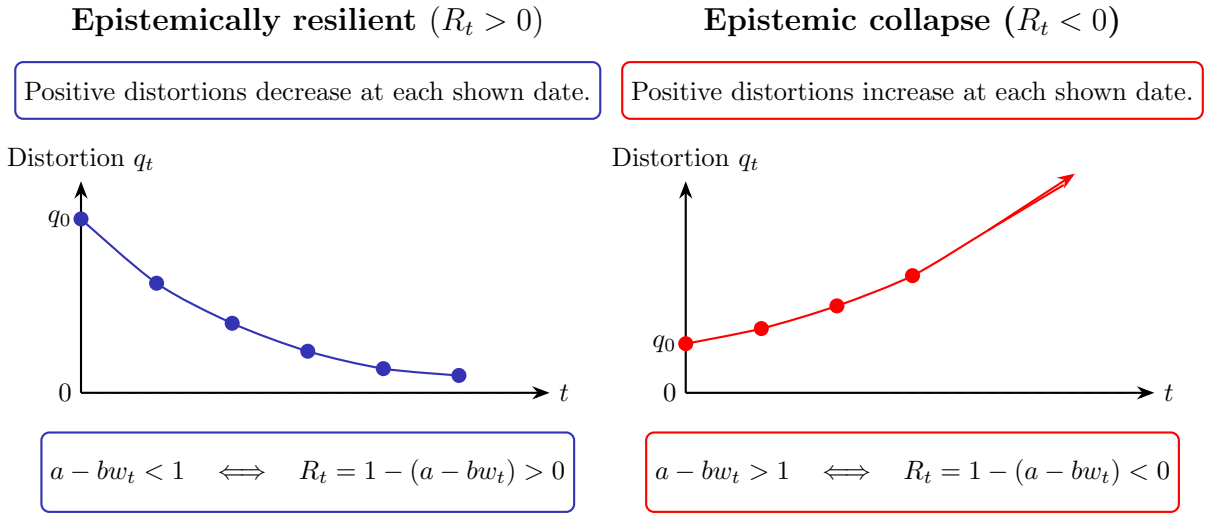


Figure 2: **Epistemic resilience and collapse.** Under $q_{t+1} = (a - bw_t)q_t = (1 - R_t)q_t$, a positive current distortion decreases over the next period when $R_t > 0$, remains unchanged when $R_t = 0$, and increases when $R_t < 0$. The illustrative paths maintain the corresponding sign at every transition shown. Subsequent correction or amplification depends on how learning efficiency evolves through $\Gamma_{\mathcal{A}}$.

Definition 4 (Loss of epistemic resilience and epistemic collapse)**.** *An AI–society system lacks epistemic resilience at date* t *when its resilience margin is non-positive:*

$$R_t \leq 0 \text{ equivalent to } w_t \leq w^\star.$$

At the boundary $R_t = 0$*, distortion remains unchanged over the next period, i.e.,* $q_{t+1} = q_t$*.*

The system experiences epistemic collapse at date t *when*

$$R_t < 0, \text{ equivalent to } w_t < w^\star,$$

Every positive current distortion is then amplified over the next period, so $q_{t+1} > q_t$ *whenever* $q_t > 0$*. If* $q_t = 0$*, then* $q_{t+1} = 0$ *regardless of* R_t*.*

efficiency may change at the next date through $\Gamma_{\mathcal{A}}$, thereby changing the resilience margin. More generally, iterating the distortion law along a feasible trajectory gives

These classifications concern the current transition. They do not by themselves imply persistence or irreversibility, because retraining can change learning efficiency and the resilience margin at the next date. Iterating the distortion law along a feasible trajectory gives

$$q_T = q_t \prod_{s=t}^{T-1} (1 - R_s), \quad T > t.$$

Whether distortion disappears or is persistently amplified therefore depends on the sequence of future margins generated by the epistemic transition.[9]

Theorem 1 provides a characterization of epistemic resilience in AI-mediated collective learning systems. At a given date, the resilience margin provides a sufficient statistic for comparing the one-period correction or amplification of a common positive informational distortion. The theory therefore shifts the object of analysis from informational distortions themselves to society's capacity to eliminate them. Consequently, the central question is no longer whether AI generates informational mistakes, but under what conditions societies retain the capacity to recover from them.

Theorem 1 characterizes epistemic resilience through the resilience margin. At a given date, this margin is a sufficient statistic for comparing the one-period correction or amplification of a common positive informational distortion. This characterization connects society's current perception rules to its capacity to eliminate errors. The epistemic transition determines how that capacity changes over time. The next subsection shows how, in the specified retraining model and under the stated conditions, this capacity can deteriorate enough to turn initial correction into sustained amplification.

2.5 Recovery and sustained amplification after a disturbance

Local asymptotic stability establishes recovery from sufficiently small feasible perturbations. It leaves open the fate of larger disturbances, including those that initially decline. We study this question in the explicit AI-retraining model of Appendix A, Section 4.2. Distortion changes the observations used for retraining. The resulting AI rule enters individual perception updating and can reduce aggregate learning efficiency. Current errors can therefore weaken the capacity to correct future errors.

Consider a disturbance $q_0 = Q \geq 0$ and consistent initial individual perceptions whose aggregate equals the distortion-free rule specified in that construction. Thus, $w_0 = \bar{w}$. Under the parameter conditions in Appendix B, Section 5.6, this initial efficiency lies above the correction threshold. The limiting efficiency generated by increasingly distorted retraining environments lies strictly below it. Every positive disturbance initially declines. Nevertheless, Theorem 2 proves that sufficiently small disturbances disappear and learning efficiency returns to $\bar{w}$, whereas sufficiently large disturbances eventually increase at every date and grow without bound under the model's autonomous dynamics.

The amplification and correction parameters remain fixed throughout. The reversal from correction to amplification comes from the decline in learning efficiency. If efficiency were held at $\bar{w}$, the same distortion law would eliminate every finite disturbance. With retraining, sustained amplification occurs for sufficiently large disturbances even though learning efficiency remains strictly positive. Every current perception rule still supports correct asymptotic learning when held fixed and applied to conditionally i.i.d. observations.

[9] Along an infinite feasible trajectory, if $R_s \geq \varepsilon$ for every $s \geq t$, with $\varepsilon \in (0, 1)$, then $q_T \leq (1 - \varepsilon)^{T-t} q_t \to 0$. If instead $R_s \leq -\varepsilon$ for every $s \geq t$, with $\varepsilon > 0$ and $q_t > 0$, then $q_T \geq (1 + \varepsilon)^{T-t} q_t \to \infty$. Strict positivity or negativity of the margin at every date, without further restrictions on its magnitude, need not imply the corresponding limit.

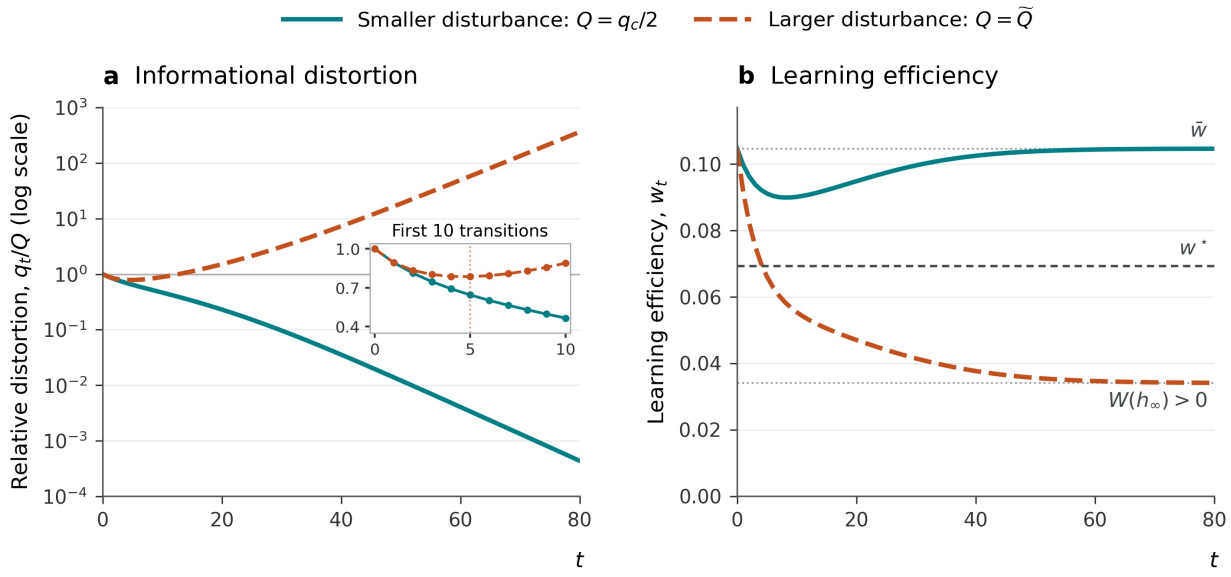


Figure 3: **Initial correction can precede sustained amplification.** Two computed trajectories differ only in initial distortion. **(a)** Relative distortion q_t/Q on a logarithmic scale. The inset shows the first ten transitions on a linear scale. **(b)** Learning efficiency and the correction threshold $w^\star$ (dashed line). Both disturbances initially decline. The larger disturbance subsequently undergoes sustained amplification as learning efficiency falls below the threshold.

Figure 3 illustrates the recovery and amplification paths. Appendix B, Section 5.6, provides the parameter values and numerical details.

These conclusions require the global conditions of the specified retraining model in addition to Assumption 1. The theorem gives sufficient disturbance bounds. It does not establish a unique threshold separating all initial disturbances or exclude recovery through an intervention. Proposition 4 in Appendix B establishes analogous conclusions for positive, strictly decreasing nonlinear distortion multipliers under the stated regularity and endpoint conditions. This extension includes sufficiently small C^1 perturbations of an affine multiplier satisfying those conditions.

3 Discussion

AI can weaken collective learning by changing how future evidence is interpreted. The theory distinguishes informational distortions circulating in society from the perception rules used to evaluate new observations. AI outputs and their circulation can affect current information quality. Retraining can transmit a distorted informational environment into the rules used to interpret subsequent evidence. The Frick-Iijima-Ishii index relates these changes in perception to aggregate learning efficiency and the capacity to correct errors. When the gross amplification parameter satisfies $a > 1$, learning efficiency can remain positive while falling below the correction threshold. Positive distortions are then amplified, which we define as epistemic collapse. This can occur while every current perception rule remains consistent and would identify the true state asymptotically if held fixed and applied to conditionally i.i.d. observations. Correct learning under each fixed rule can therefore coexist with amplification in the evolving AI–society system.

The main dynamic result shows how initial correction can precede sustained amplification. The general theory identifies the threshold for amplification at a given date and the conditions for local recovery. Theorem 2 establishes a further result for the explicit AI-retraining model. Starting from the construction's distortion-free aggregate perception rule and under the stated global conditions, every positive disturbance initially declines. Sufficiently small disturbances disappear, whereas sufficiently large disturbances eventually increase at every date and grow without bound along the autonomous trajectory. In the latter case, aggregate learning efficiency converges to a strictly positive level below the correction threshold. These trajectories use the same fixed amplification and correction parameters. Figure 3 shows how distortion can initially decline while the perception rules governing subsequent correction deteriorate.

The construction specifies how distortion changes the observations used for retraining. Population likelihood fitting determines the AI rule that enters individual perception updating. The argument allows heterogeneous individual perception rules. Proposition 4 extends the recovery and amplification results to positive, strictly decreasing nonlinear distortion multipliers under the corresponding regularity and endpoint conditions. This includes sufficiently small C^1 perturbations of an affine multiplier satisfying those conditions. The extension retains the retraining mechanism and the multiplicative dependence on current distortion. These long-run conclusions rely on global properties of the specified model in addition to Assumption 1. The sufficient disturbance bounds do not classify every intermediate disturbance or establish a unique threshold separating recovery from amplification.

The analysis adapts the broader notion of resilience as a property of a system (Holling, 1973, Walker et al., 2004) to collective knowledge production. It concerns society's capacity to recover from informational mistakes, whatever their origin. The model does not distinguish errors by source. It follows how distortion affects the interpretation of subsequent evidence and the efficiency of collective error correction.

This perspective complements approaches to AI governance centered on fairness, transparency, accountability, human oversight, and trustworthy AI (e.g., Floridi and Cowls, 2019, OECD, 2019). The recovery and amplification result shows why an initial improvement in information quality need not imply eventual recovery. An assessment based only on current content quality can miss deterioration in learning capacity before the correction threshold is crossed. The results therefore motivate attention to both dimensions. This concern is relevant to ex ante, risk-based approaches such as the European Union AI Act (European Union, 2024). Determining when and how to intervene would also require specifying intervention technologies, costs, and welfare consequences. Sustained amplification under the autonomous dynamics does not exclude recovery through an intervention.

For admissible systems under the maintained affine distortion law, comparisons use the reduced initial state and the triplet (a, b, Γ). Differences in model architecture and deployment design, including the use of transformers, retrieval-augmented generation, or mixture-of-experts models, enter through the reduced states and dynamics they generate. Systems with the same triplet on a common feasible, forward-invariant reduced state space generate identical trajectories and resilience margins from a common reduced initial condition. Their collapse thresholds coincide. At a common distortion-free steady state, their local stability properties also coincide relative to that state space. In the nonlinear extension, the corresponding transition laws are summarized by (A, Γ).

Governance interventions may change the current reduced state, the subsequent transition laws, or both. Content moderation, output filtering, and source verification may reduce current distortion or change amplification or correction effectiveness, depending on their mechanisms. Retraining protocols, data curation, model design, and institutional arrangements supporting independent interpretation may affect current per-

ception rules, the future epistemic transition, or both. Reducing current distortion alone leaves the current resilience margin unchanged under the maintained law, but can affect subsequent perception and distortion dynamics. Identifying an intervention's effect requires specifying how it enters the model. Several reduced objects may change simultaneously. For comparisons holding the post-intervention reduced initial state and b fixed, Corollary 1 identifies two components through which changes to the affine model's autonomous transition laws can affect resilience. These are distortion amplification, represented by a, and the evolution of aggregate learning efficiency, represented by Γ.

The model identifies a risk to collective learning that an initial decline in errors can conceal. Retraining can weaken correction capacity enough for those errors to become persistently amplified. Understanding how this capacity evolves, alongside current information quality, is therefore essential to assessing and governing the long-run consequences of AI-mediated interpretation.

4 Appendix A. Microfoundations of the epistemic transition

This appendix constructs the epistemic transition in two stages. Section 4.1 specifies a local response of the AI perception rule to distortion and derives the aggregate learning-efficiency transition from the individual dynamics in Equation (2). Section 4.2 then derives the AI rule from the selection of training observations and population likelihood fitting. The second construction preserves consistency on the domain used in Theorem 2 and Proposition 4. Together, the constructions show that the admissible class is nonempty and provide an explicit model for studying larger disturbances. They describe how retraining changes interpretation without specifying the internal computations of a particular AI architecture. The distortion law remains an assumption.

4.1 A local construction from perception dynamics

Consider the binary-state environment of Section 2.2. Let $z^0 \in \mathcal{C}_\mu$ be the perception rule prevailing in the absence of informational distortion, expressed as a log-likelihood-ratio vector. Let $d \in \mathbb{R}^{|X|}$ be a nonzero direction of perception change. For s in an open interval I containing zero, define

$$z(s) := z^0 + sd. \tag{9}$$

Since $\mathcal{C}_\mu$ is open, we can choose I sufficiently small that $z(s) \in \mathcal{C}_\mu$ for every $s \in I$. Every vector on this path then corresponds to a perceived signal structure with distinct full-support conditional distributions and satisfies consistency.

The scalar s measures departure from z^0 in direction d. The value $s = 0$ corresponds to the perception rule at the distortion-free steady state. This rule need not equal the true log-likelihood-ratio vector. Even in a distortion-free informational environment, collective interpretation may remain imperfect because of bounded processing, institutional conventions, or persistent model misspecification. Define the learning efficiency associated with the perception rule $z(s)$ by

$$W(s) := \omega_\mu(z(s)). \tag{10}$$

The following property follows directly from the representation in Frick et al. (2024, Eq. (10)). It identifies a class of changes in perception that leave learning efficiency unchanged.

Lemma 1 (Scale invariance of learning efficiency)**.** *For every $z \in \mathcal{Z}$ and every $c > 0$, $\omega_\mu(cz) = \omega_\mu(z)$.*

Proof. For every $z \in \mathcal{Z}$ and $c > 0$, then $cz \in \mathcal{Z}$ and $\{\nu \in \Delta(X) : \langle \nu, cz \rangle = 0\} = \{\nu \in \Delta(X) : \langle \nu, z \rangle = 0\}$. Positive rescaling leaves the feasible set in the definition of ω_μ unchanged. The two minimization problems therefore have the same value. □

The index depends on the perception vector through its indistinguishability boundary. Multiplying the vector by a positive scalar leaves that boundary unchanged. In particular, a change that only rescales the aggregate vector $\bar{z}_t$ cannot change w_t. For retraining to alter aggregate learning efficiency, it must change the indistinguishability boundary of the resulting aggregate perception rule. Therefore learning efficiency at the distortion-free rule is

$$\bar{w} := W(0) = \omega_\mu(z^0). \tag{11}$$

We impose a local condition on the effect of movement along d.

Assumption 2 (Locally efficiency-reducing perception distortion)**.** *The function W is continuously differentiable on a neighborhood of $s = 0$ and satisfies $W'(0) < 0$.*

An explicit example verifies this condition. Let $X = \{0, 1\}$ and take the true signal distributions $\mu_{\theta_L} = (0.8, 0.2)$, $\mu_{\theta_H} = (0.2, 0.8)$, where the second component is the probability of signal 1. Consider the perception path $z(s) = \big(-(0.4 - s),\, 0.6 + s\big)$. Along this path,

$$\langle \mu_{\theta_L}, z(s) \rangle = s - 0.2, \quad \langle \mu_{\theta_H}, z(s) \rangle = s + 0.4.$$

Thus, $z(s) \in \mathcal{C}_\mu$ whenever $-0.4 < s < 0.2$. Consistency holds throughout a neighborhood of zero. The indistinguishability condition $\langle \nu, z(s) \rangle = 0$ uniquely determines $\nu(1) = 0.4 - s$.

At $s = 0$, $D_{\mathrm{KL}}\big((0.6, 0.4) \| \mu_{\theta_L}\big) < D_{\mathrm{KL}}\big((0.6, 0.4) \| \mu_{\theta_H}\big)$, and this strict inequality continues to hold locally. Hence, near zero, $W(s) = D_{\mathrm{KL}}\big((0.6 + s, 0.4 - s) \| (0.8, 0.2)\big)$ and $W'(0) = -\log\left(\frac{0.4(1-0.2)}{0.2(1-0.4)}\right) = -\log\left(\frac{8}{3}\right) < 0$.

This example verifies that Assumption 2 is non-vacuous.

Scale invariance also explains why Assumption 2 excludes a purely proportional change in the perception vector. If $d = \kappa z^0$, then $z(s) = (1 + \kappa s) z^0$ and $W(s) = \bar{w}$ for all sufficiently small s, since $1 + \kappa s > 0$ there. This contradicts $W'(0) < 0$. Thus a mere change in the positive scale of a fixed perception rule cannot produce the required efficiency response. Since $W'(0) \neq 0$, the inverse function theorem gives a continuously differentiable local inverse W^{-1} on a neighborhood of $\bar{w}$.

We now specify the retraining and perception dynamics. Suppose that the AI system translates current informational distortion $q_t \in I \cap \mathbb{R}_+$ into the AI-mediated perception rule

$$z_t^A = z(q_t) = z^0 + q_t d. \tag{12}$$

Thus, in the notation of Equation (5), the retraining operator is

$$\mathcal{H}_{\mathcal{A}}(q_t, \boldsymbol{z}_t, r_t) = z(q_t).$$

This response depends only on current distortion. It has no additional internal state and does not depend on the current human perception profile, so r_t can be omitted.

Let the initial individual rules belong to $\mathcal{C}_\mu$ and suppose their aggregate satisfies perception path:

$$\bar{z}_0 = z(s_0).$$

The individual vectors $z_{i,0}$ may differ. Only their stationary network-weighted average is required to lie on the affine path. We consider paths for which $q_t \in I \cap \mathbb{R}_+$. The AI rule then remains consistent, and the convex updating rule preserves individual consistency as shown in Section 2.3. Substituting the AI-mediated perception rule $z_t^A = z(q_t)$ from Equation (12) into the aggregate perception dynamics in Equation (3) gives

$$\bar{z}_{t+1} = (1 - \lambda\rho)\bar{z}_t + \lambda\rho z(q_t).$$

If $\bar{z}_t = z(s_t)$, the affine form of z implies

If $\bar{z}_t = z(s_t)$, the affine form of z implies

$$\begin{aligned}\bar{z}_{t+1} &= (1 - \lambda\rho)z(s_t) + \lambda\rho z(q_t) \\ &= z\left((1 - \lambda\rho)s_t + \lambda\rho q_t\right).\end{aligned}$$

Since I is an interval and $\lambda\rho \in [0, 1]$, the new path parameter remains in I whenever $s_t, q_t \in I$. Thus, conditional on distortion inputs remaining in the local interval, the individual dynamics preserve the family of aggregate rules on this path. The aggregate perception parameter evolves as

$$s_{t+1} = (1 - \lambda\rho)s_t + \lambda\rho q_t. \tag{13}$$

Aggregate learning efficiency is therefore

$$w_t = \omega_\mu(\bar{z}_t) = W(s_t). \tag{14}$$

The following proposition derives the reduced epistemic transition from this perception response and individual updating.

Proposition 2 (An admissible distortion-sensitive system)**.** *Suppose Assumption 2 holds and $\lambda\rho \in (0, 1]$. Choose I as above and consider the system defined by Equations (9), (12), and (2), with initial aggregate perception $\bar{z}_0 = z(s_0)$ for $s_0 \in I$. On paths with $q_t \in I \cap \mathbb{R}_+$, the system is admissible and induces the transition*

$$\Gamma_{\mathcal{A}}(q, w) = W\left((1 - \lambda\rho)W^{-1}(w) + \lambda\rho q\right), \tag{15}$$

for $q \in I \cap \mathbb{R}_+$ and $w \in W(I)$. This transition admits a continuously differentiable extension near $(0, \bar{w})$ and satisfies $\Gamma_{\mathcal{A}}(0, \bar{w}) = \bar{w}$, $|\Gamma_{\mathcal{A},w}(0, \bar{w})| < 1$, $\Gamma_{\mathcal{A},q}(0, \bar{w}) < 0$.

Proof. Equation (14) and invertibility of W on I give $s_t = W^{-1}(w_t)$. Combining this identity with Equation (13) yields

$$\begin{aligned}w_{t+1} &= W(s_{t+1}) \\ &= W\left((1 - \lambda\rho)s_t + \lambda\rho q_t\right) \\ &= W\left((1 - \lambda\rho)W^{-1}(w_t) + \lambda\rho q_t\right).\end{aligned}$$

Next-period aggregate learning efficiency therefore depends on the perception state and current distortion

only through (q_t, w_t). This proves admissibility and Equation (15) on the stated domain. The same formula for $q \in I$ defines a continuously differentiable extension to the open set $I \times W(I)$, which contains $(0, \bar{w})$. Since $W^{-1}(\bar{w}) = 0$,

$$\Gamma_{\mathcal{A}}(0, \bar{w}) = W\left((1-\lambda\rho)W^{-1}(\bar{w})\right) = W(0) = \bar{w}.$$

Differentiating Equation (15) with respect to w gives

$$\Gamma_{\mathcal{A},w}(q, w) = W'\left((1-\lambda\rho)W^{-1}(w) + \lambda\rho q\right)(1-\lambda\rho)(W^{-1})'(w).$$

It follows that $(W^{-1})'(\bar{w}) = \frac{1}{W'(0)}$. Consequently, $\Gamma_{\mathcal{A},w}(0, \bar{w}) = W'(0)(1-\lambda\rho)\frac{1}{W'(0)} = 1-\lambda\rho$, and $\lambda\rho \in (0, 1]$ implies $|\Gamma_{\mathcal{A},w}(0, \bar{w})| = 1 - \lambda\rho < 1$. Similarly, $\Gamma_{\mathcal{A},q}(q, w) = \lambda\rho W'\left((1-\lambda\rho)W^{-1}(w) + \lambda\rho q\right)$, so $\Gamma_{\mathcal{A},q}(0, \bar{w}) = \lambda\rho W'(0) < 0$ by Assumption 2. The induced transition satisfies all the local conditions in Assumption 1. □

Equation (2). It does not assume an autonomous adjustment equation for aggregate learning efficiency. Instead, the scalar transition arises because the affine family of aggregate perception rules is invariant under the individual and AI-mediated perception dynamics, and because learning efficiency locally provides a one-to-one parametrization of this family.

Proposition 2 derives the epistemic transition from the individual dynamics in Equation (2). No separate adjustment equation for aggregate learning efficiency is imposed. The reduction follows because the aggregate perception rule remains on a one-dimensional affine path and W locally identifies its position on that path. The construction shows that the admissible class is nonempty and identifies a transparent mechanism that generates it. It does not establish a scalar Markov representation for every AI-mediated perception system.

The two derivatives separate persistence from distortion sensitivity. The derivative

$$\Gamma_w(0, \bar{w}) = 1 - \lambda\rho$$

equals the coefficient on the inherited aggregate perception parameter s_t and measures the local persistence of aggregate learning efficiency. The derivative

$$\Gamma_{\mathcal{A},q}(0, \bar{w}) = \lambda\rho W'(0)$$

measures the local effect of current distortion on next-period learning efficiency, holding current learning efficiency fixed. For a fixed perception path and learning-efficiency function W, greater responsiveness to AI interpretation, measured by $\lambda\rho$, reduces the weight on inherited aggregate perception and strengthens the negative response of next-period learning efficiency to current distortion.

4.2 AI retraining on selectively sampled data

We now construct an AI-retraining system, denoted by $\widetilde{\mathcal{A}}$, in which population likelihood fitting determines the AI perception rule. It uses the binary perception family from Section 4.1 and replaces the directly specified response $z_t^A = z(q_t)$ with $z_t^A = z(h(q_t))$. Under the parameter restriction below, the bounded response h preserves consistency for every nonnegative distortion input. This supplies the domain needed for the analysis of large disturbances. We use

$$\mu_{\theta_L} = (0.8, 0.2), \quad \mu_{\theta_H} = (0.2, 0.8) \text{ and } z(s) = (s - 0.4, s + 0.6).$$

The true signal structure remains fixed. The training data consist of state-labeled signal observations, with both labels represented. In the absence of informational distortion, their conditional distributions are

$$\begin{aligned} M^0_{\theta_L} &= (1-p_L, p_L), \quad M^0_{\theta_H} = (1-p_H, p_H), \\ p_L &= \frac{e^{0.4}-1}{e-1}, \quad p_H = e^{0.6} p_L. \end{aligned} \tag{16}$$

These probabilities satisfy $0 < p_L < p_H < 1$ and

$$\log \frac{1-p_H}{1-p_L} = -0.4, \quad \log \frac{p_H}{p_L} = 0.6.$$

Thus, fitting these training distributions gives the perception rule $z(0)$ used in the preceding construction. The training distributions may differ from the true distributions. As before, a distortion-free environment need not imply a correctly specified perception rule.

Data selection and retraining. For informational distortion $q \geq 0$, define

$$u(q) = \frac{\beta q}{1+q}, \quad \beta > 0. \tag{17}$$

An observation with signal 0 is retained with probability one. An observation with signal 1 is retained with probability $e^{-u(q)}$, regardless of its state label. Higher distortion therefore increases the underrepresentation of signal 1 in the retained data. The parameter β bounds the intensity of selection. We restrict it below to preserve consistency. Conditional on state label θ, the retained observations have distribution

$$M_\theta(q) = \left(\frac{1-p_\theta}{B_\theta(q)}, \frac{p_\theta e^{-u(q)}}{B_\theta(q)} \right), \quad B_\theta(q) = 1 - p_\theta + p_\theta e^{-u(q)}, \tag{18}$$

where $p_{\theta_L} = p_L$ and $p_{\theta_H} = p_H$. Both components are strictly positive for every $q \geq 0$. The AI fits each conditional signal distribution by minimizing the population negative log likelihood:

$$\widehat{\mu}^A_{t,\theta} = \operatorname*{arg\,min}_{p \in \Delta(X)} \left\{ -\sum_{x \in X} M_\theta(q_t)(x) \log p(x) \right\} = M_\theta(q_t). \tag{19}$$

The minimizer is unique. For any full-support candidate p, subtracting the objective's value at $p = M_\theta(q_t)$ gives $D_{\mathrm{KL}}(M_\theta(q_t) \| p)$. This difference is nonnegative and is zero only at $p = M_\theta(q_t)$. The objective is $+\infty$ if p assigns zero probability to either signal. Population fitting uses the training distribution directly and abstracts from finite-sample estimation noise. Because retention depends on the signal but not its state label, the retention factor cancels from the likelihood ratio. The normalization factors differ across states, giving

$$\log \frac{M_{\theta_H}(q)(x)}{M_{\theta_L}(q)(x)} = z(0)(x) + \log \frac{B_{\theta_L}(q)}{B_{\theta_H}(q)}, \quad x \in X.$$

The fitted AI rule is therefore

$$z^A_t = z\big(h(q_t)\big), \quad h(q) := \log \frac{B_{\theta_L}(q)}{B_{\theta_H}(q)}. \tag{20}$$

In the notation of the paper, the retraining operator is $\mathcal{H}_{\tilde{\mathcal{A}}}(q_t, \boldsymbol{z}_t, r_t) = z(h(q_t))$. It has no additional internal state and does not depend on the current human perception profile.

The response satisfies $h(0) = 0$ and

$$h'(q) = \frac{\beta e^{-u(q)}(p_H - p_L)}{(1+q)^2 B_{\theta_L}(q) B_{\theta_H}(q)} > 0. \tag{21}$$

It increases to the finite limit

$$h_\infty := \log \frac{1 - p_L + p_L e^{-\beta}}{1 - p_H + p_H e^{-\beta}} > 0. \tag{22}$$

We impose the parameter restriction

$$h_\infty < 0.2. \tag{23}$$

This restriction can be satisfied by choosing $\beta > 0$ sufficiently small, since $h_\infty \to 0$ as $\beta \downarrow 0$.

Consistency and learning efficiency. For every $s \in [0, h_\infty]$,

$$\langle \mu_{\theta_L}, z(s) \rangle = s - 0.2 < 0 < s + 0.4 = \langle \mu_{\theta_H}, z(s) \rangle.$$

Thus, $z(s) \in \mathcal{C}_\mu$ throughout this closed interval, including its endpoints. The learning-efficiency mapping from Section 2.2 is

$$W(s) := \omega_\mu(z(s)) = D_{\mathrm{KL}}\big((0.6 + s, 0.4 - s) \| (0.8, 0.2)\big), \text{ with } 0 \le s \le h_\infty. \tag{24}$$

Indeed, the unique indistinguishability distribution is $\nu_s = (0.6 + s, 0.4 - s)$, and

$$D_{\mathrm{KL}}(\nu_s \| \mu_{\theta_L}) - D_{\mathrm{KL}}(\nu_s \| \mu_{\theta_H}) = (-0.2 - 2s) \log 4 < 0.$$

Differentiating gives

$$W'(s) = \log \frac{0.2(0.6 + s)}{0.8(0.4 - s)} < 0, \text{ with } 0 \le s \le h_\infty. \tag{25}$$

Hence W is strictly decreasing and positive on this interval. Write

$$\bar{w} := W(0), \quad \underline{w} := W(h_\infty) > 0.$$

The same KL expression is smooth and strictly decreasing on $(-0.1, 0.2)$, an open interval containing $[0, h_\infty]$. The strict KL comparison above also holds on that open interval, so the expression continues to equal $\omega_\mu(z(s))$ there. Consequently, W^{-1} has a continuously differentiable extension to an open interval containing $[\underline{w}, \bar{w}]$.

Individual updating and the induced transition. Agents follow Equation (2). Let $\eta := \lambda\rho \in (0, 1]$. Suppose each initial individual perception vector belongs to $\mathcal{C}_\mu$ and

$$\bar{z}_0 = \sum_{i=1}^{n} \pi_i z_{i,0} = z(s_0) \text{ with } s_0 \in [0, h_\infty].$$

Individual rules may be heterogeneous. The following result holds for every nonnegative sequence of informational distortions. It therefore applies along feasible trajectories of the distortion law assumed in the paper.

Proposition 3 (A transition induced by data selection and retraining)**.** *Under Equations* (16) - (23) *and the*

initial conditions above, all individual and AI perception rules remain in $\mathcal{C}_\mu$ for every nonnegative distortion input sequence. The aggregate perception rule satisfies $\bar{z}_t = z(s_t)$, where

$$s_{t+1} = (1-\eta)s_t + \eta h(q_t) \text{ with } s_t \in [0, h_\infty]. \tag{26}$$

The system is admissible and induces, for $q \geq 0$ and $w \in [\underline{w}, \bar{w}]$,

$$\Gamma_{\widetilde{\mathcal{A}}}(q, w) = W\left((1-\eta)W^{-1}(w) + \eta h(q)\right). \tag{27}$$

Aggregate learning efficiency remains in $[\underline{w}, \bar{w}]$. The transition admits a continuously differentiable extension near $(0, \bar{w})$ and satisfies

$$\Gamma_{\widetilde{\mathcal{A}}}(0, \bar{w}) = \bar{w}, \Gamma_{\widetilde{\mathcal{A}},w}(0, \bar{w}) = 1-\eta, \Gamma_{\widetilde{\mathcal{A}},q}(0, \bar{w}) = -\eta\beta(p_H - p_L)\log(8/3) < 0. \tag{28}$$

In particular, the transition satisfies the local conditions of Assumption 1.

Proof. For every $q \geq 0$, Equations (20) and (23) imply $z^A = z(h(q)) \in \mathcal{C}_\mu$. Convexity of $\mathcal{C}_\mu$ and the nonnegative coefficients in Equation (2), which sum to one, then preserve individual consistency.

Stationarity of the network weights gives $\bar{z}_{t+1} = (1-\eta)\bar{z}_t + \eta z(h(q_t))$. If $\bar{z}_t = z(s_t)$, the affine form of z implies

$$\bar{z}_{t+1} = z\left((1-\eta)s_t + \eta h(q_t)\right).$$

Since $0 \leq h(q_t) < h_\infty$, induction establishes Equation (26) and preservation of the aggregate perception interval. Applying W gives $w_t = W(s_t) \in [\underline{w}, \bar{w}]$. Invertibility of W yields Equation (27), which is the factorization required for admissibility.

The formulas for u, B_θ, and h extend smoothly to a neighborhood of $q = 0$, where $1 + q$ and both normalization factors remain positive. Together with the smooth extensions of W near $s = 0$ and of W^{-1} near $\bar{w}$, this gives the stated local regularity of the transition. Finally, $W^{-1}(\bar{w}) = 0$, $h(0) = 0$, $h'(0) = \beta(p_H - p_L)$, and $W'(0) = -\log(8/3)$. Substitution and the chain rule give Equation (28). Since $0 < \eta \leq 1$, $|\Gamma_{\widetilde{\mathcal{A}},w}(0, \bar{w})| < 1$. □

This construction specifies how distortion changes the selection of training observations. The AI perception rule and its effect on learning efficiency are then derived from fitting the selected data. The negative efficiency response follows for the stated initial training distributions and selection rule. It is not a claim about arbitrary sampling distortions. Bounded selection preserves consistency and a strictly positive lower bound on aggregate learning efficiency for every nonnegative distortion input path. The construction therefore provides a domain for studying large disturbances while current perception rules remain consistent. The distortion law itself remains the assumption specified in the paper.

5 Appendix B. Proofs and further results

5.1 Error exponent of a fixed consistent perception rule

The following computation specializes the learning-efficiency argument of Frick et al. (2024) to the fixed binary-state decision rule used here. We include it to make explicit the error-exponent interpretation of our individual and aggregate perception indices.

Fix $z \in \mathcal{C}_\mu$. Let $Y_1, \ldots, Y_m$ be conditionally i.i.d. with distribution μ_θ, and let $\widehat{\theta}_m(z)$ select θ_H when $\sum_{k=1}^m z(Y_k) \geq 0$ and θ_L otherwise. Then

$$\lim_{m\to\infty} -\frac{1}{m} \log\left[\max_{\theta\in\Theta} \mathbb{P}_\theta\big(\widehat{\theta}_m(z) \neq \theta\big)\right] = \omega_\mu(z).$$

Proof. Since $z \in \mathcal{C}_\mu \subseteq \mathcal{Z}$, it is generated by a full-support perceived signal structure satisfying the consistency condition of Frick et al. (2024). Consider a binary decision problem with equal prior probabilities and payoff one for choosing the correct state and zero otherwise. Under this perceived signal structure, its subjectively optimal decision is $\widehat{\theta}_m(z)$, with ties resolved in favor of θ_H. The first-best expected payoff is one. Equation (11) of Frick et al. (2024) therefore gives

$$\frac{1}{2}\sum_{\theta\in\Theta} \mathbb{P}_\theta\big(\widehat{\theta}_m(z) \neq \theta\big) = \exp\big[-m\omega_\mu(z) + o(m)\big].$$

The maximum of the two conditional error probabilities lies between their average and twice their average. It therefore has the same exponential decay rate. □

5.2 Proof of Proposition 1.

By admissibility, $\Phi_{\mathcal{A}}(q_t, x_t) = \Gamma_{\mathcal{A}}(q_t, w_t)$. Together with the assumed distortion law $q_{t+1} = (a - bw_t)q_t$, this identity shows that the triplet $(a, b, \Gamma_{\mathcal{A}})$ determines the reduced dynamics. A feasible reduced initial condition then determines the subsequent trajectory wherever these dynamics are defined.

Now consider two admissible systems $\mathcal{A}$ and $\mathcal{B}$ such that

$$\Gamma_{\mathcal{A}} = \Gamma_{\mathcal{B}},$$

and with identical parameters a and b. Their reduced dynamical systems therefore coincide on their common reduced state space.[10] Because their common feasible reduced state space is forward invariant and their deterministic laws of motion coincide, induction gives identical trajectories (q_t, w_t) from every common reduced initial condition.

Finally, the common parameters a and b and the common learning-efficiency trajectory imply identical resilience margins $R_t = 1 - a + bw_t$ and identical epistemic collapse thresholds $w^\star = (a-1)/b$. At a common distortion-free steady state, the systems also have the same margin $\bar{R} = 1 - a + b\bar{w}$ and the same local stability properties relative to the common state space, because their reduced dynamics coincide. □

5.3 Proof of Corollary 1.

For a given post-intervention triplet (a, b, Γ), the reduced transition law is

$$(q, w) \longmapsto \big((a - bw)q, \Gamma(q, w)\big).$$

Together with a feasible reduced initial condition, this deterministic law determines the subsequent trajectory wherever the dynamics are defined.

[10] The common reduced state space is assumed feasible and forward invariant under both systems. Local stability is compared at a common distortion-free steady state. and relative to this state space.

Consider two interventions that produce the same reduced initial condition and the same triplet on their common feasible reduced state space. As in Proposition 1, induction gives identical reduced trajectories while they remain in that space. Since the parameters a and b also coincide, their resilience margins $R_t = 1-a+bw_t$ coincide at every such date.

Consequently, a difference in reduced trajectories or resilience margins requires a difference in the post-intervention initial condition, the reduced triplet, or both. Holding the initial condition fixed leaves only changes to (a, b, Γ). Holding b fixed as well leaves only a and Γ. These conditions are necessary for an effect. A change to one component may have no effect on a particular trajectory. □

5.4 Proof of Theorem 1.

For a small positive informational distortion, the distortion law implies $\frac{q_{t+1}}{q_t} = a - bw_t$. Because we restrict attention to the state space on which $a - bw_t \geq 0$, the distortion is self-correcting at date t if and only if $a - bw_t < 1$.

At the distortion-free learning efficiency $w_t = \bar{w}$, this condition becomes $a - b\bar{w} < 1$, or equivalently, $\bar{R} := 1 - (a - b\bar{w}) > 0$. Since $b > 0$, this is also equivalent to $\bar{w} > \frac{a-1}{b} = w^\star$.

Because the mapping $w \mapsto a - bw$ is continuous, whenever $\bar{R} > 0$ there exists a neighborhood of $\bar{w}$ in which $a - bw < 1$. Hence sufficiently small informational distortions remain self-correcting for feasible states sufficiently close to the distortion-free steady state. Conversely, if $\bar{R} \leq 0$, then at $a - b\bar{w} \geq 1$, so a small positive distortion does not decrease. The system is therefore not locally epistemically resilient.

Finally, at their respective distortion-free learning efficiencies, systems A and B have distortion multipliers

$$a_{\mathcal{A}} - b_{\mathcal{A}}\bar{w}_{\mathcal{A}} \text{ and } a_{\mathcal{B}} - b_{\mathcal{B}}\bar{w}_{\mathcal{B}}.$$

For a common positive distortion, system A produces no greater next-period distortion than system B if and only if

$$a_{\mathcal{A}} - b_{\mathcal{A}}\bar{w}_{\mathcal{A}} \leq a_{\mathcal{B}} - b_{\mathcal{B}}\bar{w}_{\mathcal{B}}.$$

This is equivalent to $\bar{R}_{\mathcal{A}} \geq \bar{R}_{\mathcal{B}}$. □

Theorem 1 concerns correction over one period. The following corollary relates its strict inequality to local asymptotic stability of the joint reduced system under Assumption 1 and explicit conditions on the feasible state space.

5.5 Corollary 2.

Corollary 2 (Local stability of the distortion-free steady state)**.** *Suppose Assumption 1 holds and $a-b\bar{w} \geq 0$. Let $\mathcal{D}$ be a feasible reduced state space on which the reduced equations describe the induced dynamics, with*

$$(0, \bar{w}) \in \mathcal{D} \subseteq \left\{(q, w) \in \mathbb{R}^2_+ : a - bw \geq 0\right\}.$$

Assume that $\mathcal{D}$ is locally forward invariant and satisfies

$$(q, \bar{w}) \in \mathcal{D} \text{ for all sufficiently small } q \geq 0.$$

This condition permits small perturbations in informational distortion while holding learning efficiency at $\bar{w}$. Then the distortion-free steady state $(0, \bar{w})$ is locally asymptotically stable relative to $\mathcal{D}$ if and only if $\bar{R} := 1 - (a - b\bar{w}) > 0$. If $\bar{R} \leq 0$, it is unstable relative to $\mathcal{D}$.

Proof. Write the reduced map as

$$F(q, w) = ((a - bw)q,\, \Gamma_{\mathcal{A}}(q, w))\,.$$

Let $c := \Gamma_{\mathcal{A},q}(0, \bar{w})$ and $d := \Gamma_{\mathcal{A},w}(0, \bar{w})$. Assumption 1 gives $c < 0$ and $|d| < 1$. The consistency restriction implies $\bar{w} > 0$.

Write the reduced map as $F(q, w) = ((a - bw)q,\, \Gamma_{\mathcal{A}}(q, w))$. Let $c := \Gamma_{\mathcal{A},q}(0, \bar{w}) < 0$ and $d := \Gamma_{\mathcal{A},w}(0, \bar{w})$. Assumption 1 gives $c < 0$ and $|d| < 1$. The consistency restriction implies $\bar{w} > 0$. The Jacobian at the distortion-free steady state is

$$DF(0, \bar{w}) = \begin{pmatrix} a - b\bar{w} & 0 \\ c & d \end{pmatrix}.$$

It is triangular, with eigenvalues $\lambda_1 = a - b\bar{w} = 1 - \bar{R}$ and $\lambda_2 = d$.

Case 1. $\bar{R} > 0$. Since $a - b\bar{w} \geq 0$, this case implies $0 \leq \lambda_1 < 1$. Together with $|d| < 1$, both eigenvalues lie strictly inside the unit disk. Differentiability therefore implies that, in a suitable equivalent norm, there exist $\gamma \in (0, 1)$ and a neighborhood of $(0, \bar{w})$ such that $\|F(q, w) - (0, \bar{w})\| \leq \gamma \|(q, w) - (0, \bar{w})\|$ for states in that neighborhood. Local forward invariance of $\mathcal{D}$ then implies local asymptotic stability relative to $\mathcal{D}$.

Case 2. $\bar{R} < 0$. Here $a - b\bar{w} > 1$. Choose $\delta > 0$ sufficiently small that $a - b\bar{w} - b\delta > 1$. As long as a trajectory remains in the neighborhood $q < \delta$ and $|w - \bar{w}| < \delta$, its distortion satisfies

$$q_{t+1} = (a - bw_t)q_t \geq (a - b\bar{w} - b\delta)q_t.$$

Starting from any sufficiently small feasible perturbation $(q_0, \bar{w})$ with $q_0 > 0$, the trajectory cannot remain in this neighborhood indefinitely, because the displayed lower bound would force q_t to grow geometrically. Such initial conditions can be chosen arbitrarily close to $(0, \bar{w})$, establishing instability relative to $\mathcal{D}$.

Case 3. $\bar{R} = 0$. In this case $a - b\bar{w} = 1$, so the first eigenvalue equals one and linearization alone does not determine stability. Set $y = w - \bar{w}$. The translated dynamics are

$$q' = q - bqy \text{ and } y' = cq + dy + r(q, y),$$

where differentiability gives $r(q, y) = o(q + |y|)$ as $(q, y) \to (0, 0)$, $q \geq 0$. Define $\alpha := \frac{c}{1-d} < 0$. Choose $\varepsilon \in (0, -\alpha)$ and consider the cone

$$\mathcal{K}_\varepsilon := \{(q, y) : q > 0,\ |y - \alpha q| \leq \varepsilon q\}\,.$$

We first show that trajectories in this cone remain in it while sufficiently close to the origin. For $(q, y) \in \mathcal{K}_\varepsilon$, write $v := \frac{y}{q} \in [\alpha - \varepsilon, \alpha + \varepsilon]$. Since $\alpha + \varepsilon < 0$, $q' = q(1 - bqv) > 0$. Moreover,

$$\frac{y'}{q'} = \frac{c + dv + r(q, qv)/q}{1 - bqv} = \alpha + d(v - \alpha) + o(1)$$

as $q \downarrow 0$, uniformly for $v \in [\alpha - \varepsilon, \alpha + \varepsilon]$. Uniformity follows from $r(q, y) = o(q + |y|)$ and boundedness of

this interval. Because $|d| < 1$, for all sufficiently small $q > 0$,

$$\left| \frac{y'}{q'} - \alpha \right| < \varepsilon.$$

Thus the image remains in $\mathcal{K}_\varepsilon$. Within this cone, $q' - q = -bq^2 v \geq -b(\alpha+\varepsilon)q^2 = \kappa q^2$ and $\kappa := -b(\alpha+\varepsilon) > 0$. Choose $\delta > 0$ sufficiently small that the preceding estimates hold throughout $\mathcal{K}_\varepsilon \cap \{q < \delta\}$. A trajectory entering this truncated cone cannot remain there forever. Indeed, if it entered at date N and remained there, then $q_t \geq q_N > 0$ for all $t \geq N$ and

$$q_t \geq q_N + (t - N)\kappa q_N^2,$$

contradicting $q_t < \delta$. It must therefore reach $q_t \geq \delta$ in finite time.

It remains to show that feasible perturbations of the form $(q_0, \bar{w})$ enter the cone. Choose a fixed integer $N \geq 1$ such that $|\alpha|\,|d|^N < \frac{\varepsilon}{2}$. The derivative of the N-fold iterate of the translated map at the origin is

$$\begin{pmatrix} 1 & 0 \\ c & d \end{pmatrix}^N = \begin{pmatrix} 1 & 0 \\ \alpha(1 - d^N) & d^N \end{pmatrix}.$$

Consequently, for initial conditions $q_0 > 0$ and $y_0 = 0$,

$$q_N = q_0 + o(q_0), \quad y_N = \alpha(1 - d^N)q_0 + o(q_0)$$

as $q_0 \downarrow 0$. Hence $\dfrac{y_N}{q_N} \longrightarrow \alpha(1 - d^N)$. By the choice of N, all sufficiently small such perturbations enter $\mathcal{K}_\varepsilon$ with $0 < q_N < \delta$. Their first N iterates remain arbitrarily close to the steady state as $q_0 \downarrow 0$, by continuity. The initial conditions belong to $\mathcal{D}$ by assumption, and local forward invariance ensures feasibility while the trajectory remains in the local neighborhood. The subsequent cone dynamics force these trajectories to leave a fixed neighborhood of the steady state. Since q_0 can be arbitrarily small, $(0, \bar{w})$ is unstable relative to $\mathcal{D}$ at $\bar{R} = 0$.

Combining the three cases establishes the result. □

5.6 Recovery and sustained amplification

Local asymptotic stability guarantees recovery from sufficiently small perturbations. To examine larger disturbances, we use the AI-retraining model $\widetilde{\mathcal{A}}$ constructed in Appendix A, Section 4.2. Informational distortion changes which observations are selected for retraining. Population likelihood fitting of their conditional signal distributions gives the AI perception rule $z_t^A = z(h(q_t))$. Here $z(s)$ is the affine perception family specified in that construction, and $h(0) = 0$, $h'(q) > 0$, and $h(q) \uparrow h_\infty < 0.2$ as $q \to \infty$. The aggregate perception rule is $\bar{z}_t = z(s_t)$, where s_t indexes distortion in interpretation and remains in $[0, h_\infty]$. Learning efficiency is $w_t = W(s_t)$, with $W(s) := \omega_\mu(z(s))$ and $W'(s) < 0$ on this interval. In particular,

$$\bar{w} = W(0) > \underline{w} = W(h_\infty) > 0.$$

Under the initial conditions of Proposition 3, all individual and AI perception rules remain consistent. Combining the aggregate perception dynamics derived in that proposition with the assumed distortion law

gives

$$\begin{aligned} q_{t+1} &= (a - bW(s_t))q_t, \\ s_{t+1} &= (1-\eta)s_t + \eta h(q_t), \quad \eta = \lambda\rho \in (0,1]. \end{aligned} \tag{29}$$

Since W is strictly decreasing, (q_t, s_t) is an equivalent parametrization of the reduced state (q_t, w_t) for this model. Consider the parameter regime

$$0 < m_0 := a - b\bar{w} < 1 < m_\infty := a - b\underline{w}, \quad a, b > 0. \tag{30}$$

At the distortion-free learning efficiency $\bar{w}$, positive distortions decline. At the limiting efficiency $\underline{w}$ generated by increasingly distorted training environments, they increase. Equivalently, the correction threshold satisfies $\underline{w} < w^\star < \bar{w}$, and the multiplier m_0 is strictly positive. A choice of parameters satisfying these conditions is given below. Define

$$s_c := W^{-1}(w^\star), \quad q_c := h^{-1}(s_c), \quad w^\star = \frac{a-1}{b}. \tag{31}$$

Then $0 < s_c < h_\infty$ and $q_c > 0$. The pair (q_c, s_c) is the unique positive steady state in (q, s) coordinates.

Theorem 2 (Recovery and sustained amplification)**.** *Consider the AI-retraining model (Proposition 3), with parameters satisfying Condition (30) and consistent initial individual perceptions whose aggregate is $z(0)$. Let $s_0 = 0$ and $q_0 = Q \geq 0$, representing a disturbance at date zero followed by the autonomous dynamics in Equation (29). The distortion-free reduced steady state $(0, \bar{w})$ is locally asymptotically stable relative to the feasible reduced state space. Every positive disturbance initially declines,*

$$q_1 = m_0 Q < Q \ \text{ for every } Q > 0.$$

Its subsequent evolution satisfies the following conclusions.

1. *If $0 \leq Q \leq q_c$, then $q_t \longrightarrow 0$, $s_t \longrightarrow 0$, $w_t \longrightarrow \bar{w}$.*
2. *There exists a finite $\widetilde{Q} > q_c$ such that, for every $Q \geq \widetilde{Q}$, $q_t \longrightarrow \infty$, $s_t \longrightarrow h_\infty$, $w_t \longrightarrow \underline{w} > 0$. Moreover, $q_{t+1} > q_t$ at every sufficiently late date, and $\frac{q_{t+1}}{q_t} \longrightarrow m_\infty > 1$.*

All trajectories remain feasible, all individual and AI perception rules remain in $\mathcal{C}_\mu$, and $w_t \in [\underline{w}, \bar{w}]$ throughout.

Proof. Define $m(s) := a - bW(s)$ and $T(q,s) := \big(m(s)q, (1-\eta)s + \eta h(q)\big)$. On $[0, h_\infty]$, the function m is continuous and strictly increasing, with $m(0) = m_0 \in (0,1)$ and $m(h_\infty) = m_\infty > 1$. Since $0 \leq h(q) < h_\infty$ for every finite $q \geq 0$, the state space $\mathcal{D}_s := \mathbb{R}_+ \times [0, h_\infty]$ is forward invariant under T. In particular,

$$m_0 q_t \leq q_{t+1} \leq m_\infty q_t. \tag{32}$$

Thus all trajectories are defined, finite, and feasible at every finite date, and $q_t > 0$ whenever $q_0 > 0$. Consistency of the individual and AI perception rules and the efficiency bounds follow from Proposition 3.

Local stability and the positive steady state. At $(q, s) = (0, 0)$, the Jacobian is

$$DT(0,0) = \begin{pmatrix} m_0 & 0 \\ \eta h'(0) & 1-\eta \end{pmatrix}.$$

Its eigenvalues m_0 and $1-\eta$ lie strictly inside the unit disk. The continuously differentiable extensions established in the retraining construction therefore imply local asymptotic stability relative to $\mathcal{D}_s$. The change of variables $w = W(s)$ is a continuously differentiable local diffeomorphism near $s = 0$. Consequently, the corresponding reduced steady state $(0, \bar{w})$ is locally asymptotically stable in (q, w) coordinates. For $s_0 = 0$, the initial transition is $q_1 = m_0 Q < Q$ whenever $Q > 0$.

Condition (30) and strict monotonicity of m yield a unique $s_c \in (0, h_\infty)$ with $m(s_c) = 1$. The continuous, strictly increasing function h maps $[0, \infty)$ onto $[0, h_\infty)$. Hence a unique finite $q_c > 0$ satisfies $h(q_c) = s_c$, and (q_c, s_c) is a steady state. Conversely, any steady state with $q > 0$ must satisfy $m(s) = 1$ and, since $\eta > 0$, $s = h(q)$. It must therefore equal (q_c, s_c).

A convergence property of the perception update. We will use the following consequence of the second equation in (29). Whenever $h(q_t) \to v$,

$$s_t \longrightarrow v. \tag{33}$$

To establish this property, fix $\varepsilon > 0$ and choose t_0 such that $|h(q_t) - v| \le \varepsilon$ for every $t \ge t_0$. Iteration gives, for $k \ge 1$, $|s_{t_0+k} - v| \le (1-\eta)^k |s_{t_0} - v| + \varepsilon\big(1 - (1-\eta)^k\big)$. Taking the limit superior and then letting $\varepsilon \downarrow 0$ proves (33), including when $\eta = 1$.

An invariant recovery region. Let $\mathcal{L} := [0, q_c] \times [0, s_c]$. For $(q, s) \in \mathcal{L}$, monotonicity gives $m(s) \le 1$ and $h(q) \le s_c$. Thus,

$$0 \le m(s) q \le q \le q_c, \quad 0 \le (1-\eta) s + \eta h(q) \le s_c.$$

The region $\mathcal{L}$ is therefore forward invariant, and distortion is nonincreasing there. Every trajectory starting in $\mathcal{L} \setminus \{(q_c, s_c)\}$ has $q_j < q_c$ at some finite date j. This holds initially if $q_0 < q_c$. Otherwise, $s_0 < s_c$ implies $q_1 = m(s_0) q_c < q_c$. Consequently, q_t converges to a limit $\ell \in [0, q_j]$, with $\ell < q_c$. Continuity of h and Equation (33) imply $s_t \to h(\ell) < s_c$. Therefore $m(s_t)$ is eventually bounded above by a constant strictly below one, which forces $q_t \to 0$. It follows that $s_t \to h(0) = 0$ and $w_t \to W(0) = \bar{w}$. Every initial state $(Q, 0)$ with $0 \le Q \le q_c$ belongs to this recovery region and differs from (q_c, s_c). This proves part 1.

An invariant amplification region. Let $\mathcal{U} := [q_c, \infty) \times [s_c, h_\infty]$. For $(q, s) \in \mathcal{U}$, we have $m(s) \ge 1$ and $h(q) \ge s_c$, so $\mathcal{U}$ is forward invariant and distortion is nondecreasing there. Every trajectory starting in $\mathcal{U} \setminus \{(q_c, s_c)\}$ has $q_j > q_c$ at some finite date j. Either this holds initially, or $q_0 = q_c$ and $s_0 > s_c$ imply $q_1 > q_c$. If q_t were bounded, it would converge to a finite $\ell \ge q_j > q_c$. Equation (33) would then give $s_t \to h(\ell) > s_c$. The multiplier $m(s_t)$ would eventually be bounded below by a constant strictly above one, contradicting boundedness of q_t. It follows that $q_t \to \infty$. Consequently, $h(q_t) \to h_\infty$, and Equation (33) implies

$$s_t \to h_\infty, \quad w_t \to \underline{w} > 0 \text{ and } \frac{q_{t+1}}{q_t} = m(s_t) \to m_\infty > 1.$$

Entry after a sufficiently large initial disturbance. Since $h(2q_c) > s_c$ and $\eta > 0$, choose a finite integer $N \ge 1$ such that $\big(1 - (1-\eta)^N\big) h(2q_c) > s_c$. For $\eta = 1$, $N = 1$ suffices. Set beginequation $\widetilde{Q} := \frac{2q_c}{m_0^N} > q_c$. For any $Q \ge \widetilde{Q}$, the lower bound in (32) gives $q_k \ge m_0^k Q \ge 2q_c$ for $0 \le k \le N$. Starting from $s_0 = 0$, iteration

of the perception update gives

$$s_N = \eta \sum_{j=0}^{N-1} (1-\eta)^{N-1-j} h(q_j) \geq \big(1-(1-\eta)^N\big) h(2q_c) > s_c.$$

Thus $q_N \geq 2q_c > q_c$ and $s_N > s_c$. The trajectory has entered $\mathcal{U}$ strictly above both coordinates of its steady state. It remains strictly above those coordinates thereafter, so $q_{t+1} > q_t$ for every $t \geq N$. The amplification-region argument supplies all the remaining conclusions in part 2. □

Nonemptiness of the parameter regime. For any selection parameter β satisfying $0 < h_\infty < 0.2$, choose any $b \in \left(0, \dfrac{2}{\bar{w}-\underline{w}}\right)$ and let $a := 1 + \frac{b}{2}(\bar{w}+\underline{w})$. Then,

$$m_0 = 1 - b\frac{\bar{w}-\underline{w}}{2} \in (0,1) \text{ and } m_\infty = 1 + b\frac{\bar{w}-\underline{w}}{2} > 1.$$

Thus Condition (30) can be satisfied. Its strict inequalities persist under sufficiently small changes in these parameters. The sufficient bound in Equation (5.6) need not be minimal. The proof does not classify all initial conditions outside the two invariant regions.

Numerical illustration in Figure 3. Figure 3 compares two trajectories of Equation (29), both starting from $s_0 = 0$ and hence $w_0 = \bar{w}$. We set $\beta = 0.4$, $\eta = 0.2$, $b = 3$, and $a = 1 + \frac{b}{2}(\bar{w}+\underline{w}) \simeq 1.208057$. The initial disturbances are $Q = q_c/2 \simeq 0.370180$ and $Q = \widetilde{Q} \simeq 2.898220$, using $N = 6$ in Equation (5.6).These choices satisfy the respective sufficient conditions of Theorem 2, which establishes their long-run outcomes.

For the smaller disturbance, learning efficiency remains above $w^\star$ and the system recovers. For the larger disturbance, distortion reaches its minimum at $t = 5$, marked by the vertical dotted line in the inset, and increases at every subsequent transition. Points in the inset mark the discrete-time iterates. The horizontal dotted lines in panel (b) mark $\bar{w}$ and $\underline{w} = W(h_\infty) > 0$. Normalization in panel (a) affects only the display, and retraining responds to the unnormalized distortion q_t.

5.7 Robustness to a nonlinear distortion multiplier

Retain the AI-retraining model of Proposition 3, including $\eta \in (0,1]$, the perception response h, and the learning-efficiency mapping W. Write $\mathcal{J} = [\underline{w}, \bar{w}]$, where $\bar{w} = W(0) > \underline{w} = W(h_\infty) > 0$. Replace the distortion law by

$$q_{t+1} = A(w_t) q_t. \tag{34}$$

The function $A : J \to (0,\infty)$ is continuously differentiable on an open interval J containing $\mathcal{J}$, strictly decreasing on $\mathcal{J}$, and satisfies

$$0 < m_{0,A} := A(\bar{w}) < 1 < m_{\infty,A} := A(\underline{w}). \tag{35}$$

The training-data selection, retraining, and perception updates are unchanged. The induced epistemic transition therefore remains the one in Equation (27). In (q,s) coordinates, the coupled dynamics become

$$\begin{aligned} q_{t+1} &= A(W(s_t)) q_t, \\ s_{t+1} &= (1-\eta) s_t + \eta h(q_t). \end{aligned} \tag{36}$$

There is a unique $w_A^\star \in (\underline{w}, \bar{w})$ such that $A(w_A^\star) = 1$. Define

$$s_{c,A} := W^{-1}(w_A^\star), \quad q_{c,A} := h^{-1}(s_{c,A}).$$

Then $0 < s_{c,A} < h_\infty$ and $q_{c,A} > 0$. The instantaneous resilience margin for this extension is $R_t^A := 1 - A(w_t)$. For $q_t > 0$, distortion declines, remains unchanged, or increases between dates t and $t+1$ according as $w_t > w_A^\star$, $w_t = w_A^\star$, or $w_t < w_A^\star$. For the affine multiplier, $R_t^A = R_t$ and $w_A^\star = (a-1)/b$.

Proposition 4 (Recovery and amplification with a nonlinear multiplier)**.** *Under Condition (35), consider consistent initial individual perceptions whose aggregate is $z(0)$, with $s_0 = 0$ and $q_0 = Q \geq 0$. The distortion-free reduced steady state $(0, \bar{w})$ is locally asymptotically stable relative to the feasible reduced state space, and*

$$q_1 = m_{0,A} Q < Q \ \text{ for every } Q > 0.$$

The pair $(q_{c,A}, s_{c,A})$ is the unique positive steady state in (q, s) coordinates. The following conclusions hold:

1. *If $0 \leq Q \leq q_{c,A}$, then $q_t \to 0$, $s_t \to 0$, $w_t \to \bar{w}$.*
2. *Choose any integer $N_A \geq 1$ satisfying $\big(1 - (1-\eta)^{N_A}\big) h(2q_{c,A}) > s_{c,A}$, and define the finite sufficient bound $\widetilde{Q}_A := \frac{2q_{c,A}}{m_{0,A}^{N_A}} > q_{c,A}$. For every $Q \geq \widetilde{Q}_A$,*

$$q_t \to \infty, \quad s_t \to h_\infty, \quad w_t \to \underline{w} > 0 \text{ and } q_{t+1} > q_t for all t \geq N_A \text{ with } \frac{q_{t+1}}{q_t} \to m_{\infty,A} > 1.$$

All trajectories remain feasible, all individual and AI perception rules remain consistent, and $w_t \in [\underline{w}, \bar{w}]$ throughout.

Proof. Set $m_A(s) := A(W(s))$. Since both A and W are strictly decreasing on the relevant intervals, m_A is continuous and strictly increasing on $[0, h_\infty]$. Its endpoint values are $m_{0,A} \in (0,1)$ and $m_{\infty,A} > 1$. The state space $\mathbb{R}_+ \times [0, h_\infty]$ is forward invariant under Equation (36), and

$$m_{0,A} q_t \leq q_{t+1} \leq m_{\infty,A} q_t.$$

Hence trajectories are feasible and finite at every finite date, and positive initial distortion remains positive. Consistency and the efficiency bounds follow from Proposition 3. The Jacobian in (q, s) coordinates at $(0,0)$ is

$$\begin{pmatrix} m_{0,A} & 0 \\ \eta h'(0) & 1-\eta \end{pmatrix}.$$

Both eigenvalues have modulus strictly less than one. The continuously differentiable local extension and the local change of variables $w = W(s)$ therefore establish local asymptotic stability of $(0, \bar{w})$ relative to the feasible reduced state space. The initial decline follows from $s_0 = 0$. A positive steady state must satisfy $m_A(s) = 1$ and $s = h(q)$. Strict monotonicity gives precisely $(q_{c,A}, s_{c,A})$.

The perception update is unchanged, so the convergence property in Equation (33) continues to hold. If $h(q_t) \to v$, then $s_t \to v$. The rectangle

$$\mathcal{L}_A := [0, q_{c,A}] \times [0, s_{c,A}]$$

is forward invariant, and q_t is nonincreasing there. Starting at $(Q, 0)$ with $Q \leq q_{c,A}$, we have $q_j < q_{c,A}$ for

some $j \in \{0,1\}$. Indeed, this holds initially if $Q < q_{c,A}$, and $q_1 = m_{0,A} q_{c,A} < q_{c,A}$ when $Q = q_{c,A}$. Thus $q_t \to \ell < q_{c,A}$ and $s_t \to h(\ell) < s_{c,A}$. The multiplier $m_A(s_t)$ is eventually bounded above by a constant strictly below one. Consequently, $q_t \to 0$, $s_t \to 0$, and $w_t \to \bar{w}$, proving part 1.

Likewise, the region

$$\mathcal{U}_A := [q_{c,A}, \infty) \times [s_{c,A}, h_\infty]$$

is forward invariant, and q_t is nondecreasing there. Any trajectory entering this region with $q > q_{c,A}$ must satisfy $q_t \to \infty$. Otherwise, it would converge to a finite $\ell > q_{c,A}$, implying $s_t \to h(\ell) > s_{c,A}$. The multiplier would then eventually be bounded below by a constant strictly above one, contradicting boundedness of q_t. The perception convergence property therefore yields $s_t \to h_\infty$, $w_t \to \underline{w}$, and $q_{t+1}/q_t \to m_{\infty,A}$.

Since $h(2q_{c,A}) > s_{c,A}$ and $\eta > 0$, a finite N_A satisfying the stated condition exists. When $\eta = 1$, take $N_A = 1$. If $Q \geq \widetilde{Q}_A$, the multiplier bound gives

$$q_k \geq m_{0,A}^k Q \geq m_{0,A}^{N_A} Q \geq 2q_{c,A} \text{ for } 0 \leq k \leq N_A.$$

Starting from $s_0 = 0$, iteration gives

$$s_{N_A} = \eta \sum_{j=0}^{N_A - 1} (1-\eta)^{N_A - 1 - j} h(q_j) \geq \big(1 - (1-\eta)^{N_A}\big) h(2q_{c,A}) > s_{c,A}.$$

Thus the trajectory enters $\mathcal{U}_A$ strictly above both coordinates of its steady state at date N_A and remains strictly above them thereafter. This proves part 2, including strict amplification for every $t \geq N_A$. □

A nonlinear specification and small perturbations. For example, consider $A(w) = \exp\big(\kappa(w^\dagger - w)\big)$, $\kappa > 0$ and $w^\dagger \in (\underline{w}, \bar{w})$. This globally positive nonlinear multiplier satisfies all the conditions, with $w_A^\star = w^\dagger$. The conclusions also hold for all sufficiently small C^1 perturbations of an affine multiplier satisfying Condition (30). To see this, let $A_0(w) = a - bw$ satisfy that condition, and set

$$\delta_0 := \frac{1}{2} \min\{m_0,\, 1 - m_0,\, m_\infty - 1\} > 0.$$

Consider any continuously differentiable $\widetilde{A}$ defined on an open interval containing $\mathcal{J}$ and satisfying

$$\sup_{w \in \mathcal{J}} |\widetilde{A}(w) - A_0(w)| < \delta_0 \text{ and } \sup_{w \in \mathcal{J}} |\widetilde{A}'(w) + b| < \frac{b}{2},$$

These bounds imply $\widetilde{A} > 0$ and $\widetilde{A}' < -b/2 < 0$ on $\mathcal{J}$, together with $\widetilde{A}(\bar{w}) < 1 < \widetilde{A}(\underline{w})$. By continuity and compactness of $\mathcal{J}$, the domain can be restricted to an open interval containing $\mathcal{J}$ on which $\widetilde{A}$ is positive. Proposition 4 therefore applies, with the threshold and sufficient disturbance bounds computed from $\widetilde{A}$.

This extension retains the multiplicative dependence on current distortion and the specified AI-retraining mechanism. It establishes robustness to changes in how learning efficiency affects the distortion multiplier, subject to the stated conditions. As in Theorem 2, the sufficient bounds do not classify every intermediate disturbance or establish a unique dividing threshold for its size. Sustained amplification describes the autonomous dynamics and does not rule out recovery following an intervention.

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